SET 74 • HSF-IF.C.7.b

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\sqrt{81}$ and $\sqrt[3]{(-27)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q1/5

1602

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9 and −3

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = 125$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q2/5

1603

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\sqrt[3]{8} + \sqrt{4}$?

A. 6

B. 12

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q3/5

1604

QUESTION ID

S74-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

GRADING NOTE 2 + 2 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{0.25} = 0.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q4/5

1605

QUESTION ID

S74-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.5^2 = 0.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt[3]{(-8)}$ is not a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q5/5

1606

QUESTION ID

S74-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt[3]{(-8)} = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check the solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve √(x + 4) = 3.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q1/5

1608

QUESTION ID

S74-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. x + 4 = 9. Check: √9 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sqrt[3]{(2x)} = -2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q2/5

1609

QUESTION ID

S74-F1-Q2

TYPE

Short answer

CORRECT ANSWER

−4

GRADING NOTE Any equivalent form. 2x = −8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q2/5

Solve $\sqrt{(x + 2)} = x$.

A. $x = -1$ only

B. $x = 2$ or $x = -1$

C. No solution

D. $x = 2$ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q3/5

1610

QUESTION ID

S74-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 only

GRADING NOTE $x^2 - x - 2 = 0$ gives 2 or −1; $x = -1$ is extraneous because $\sqrt{1} = 1$, not −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(x - 3)} = -2$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q4/5

1611

QUESTION ID

S74-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A principal square root is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q4/5

True or false?

Cubing both sides of $\sqrt[3]{x} = 4$ gives $x = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q5/5

1612

QUESTION ID

S74-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.b

FORWARD 2

Find inverses of square root and cube root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{x}$ for $x \geq 0$. Find $f^{-1}(5)$.

$f^{-1}(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q1/5

1614

QUESTION ID

S74-F2-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form. $\sqrt{25} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = \sqrt[3]{x}$. Find $g^{-1}(-2)$.

$g^{-1}(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q2/5

1615

QUESTION ID

S74-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

GRADING NOTE Any equivalent form. $\sqrt[3]{(-8)} = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the inverse of $f(x) = \sqrt{(x - 1)}$, for $x \geq 1$?

A. $f^{-1}(x) = \sqrt{(x + 1)}$

B. $f^{-1}(x) = x^2 + 1$, for $x \geq 0$

C. $f^{-1}(x) = x^2 - 1$

D. $f^{-1}(x) = (x + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q3/5

1616

QUESTION ID

S74-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f^{-1}(x) = x^2 + 1$, for $x \geq 0$

GRADING NOTE $y = \sqrt{(x - 1)}$ gives $x = y^2 + 1$ with $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q3/5

True or false?

The graph of $y = \sqrt{x}$ $(x \geq 0)$ is the reflection of the graph of $y = x^2$ $(x \geq 0)$ across the line $y = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q4/5

1617

QUESTION ID

S74-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are inverse functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$g(x) = \sqrt[3]{x}$ has no inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q5/5

1618

QUESTION ID
S74-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its inverse is $g^{-1}(x) = x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and step functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate piecewise-defined functions	**HSF-IF.A.2**
BACKWARD 2	Order numbers and describe intervals on a number line	**6.NS.C.7.a**
FORWARD 1	Graph the probability distribution of a discrete random variable —..	**HSS-MD.A.1**
FORWARD 2	Read inverse values of piecewise and step functions	**HSF-BF.B.4.c**

SET 75 • HSF-IF.C.7.b

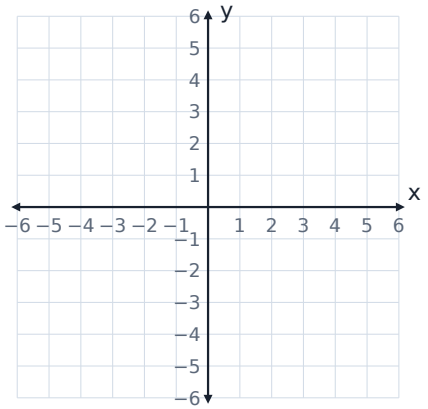
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 1$ for $x < 2$, and
$f(x) = 5 - x$ for $x \geq 2$.

Graph f. Then find f(0), f(2), and f(4).



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q1/5

1621

QUESTION ID

S75-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A line rising to (2, 3) from the left, joined at (2, 3) to a line falling to the right through (4, 1); (2, 3) is a filled point

Written: f(0) = 1, f(2) = 3, f(4) = 1

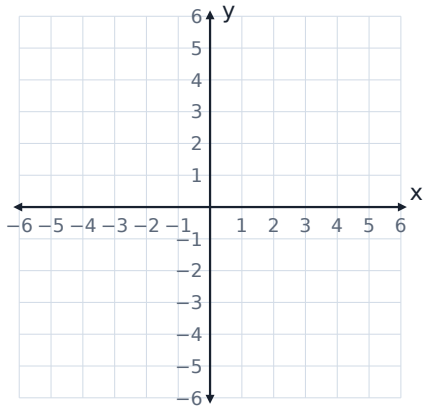
GRADING NOTE Form required: all three values. Both the graph and the values are graded. x = 2 belongs to the second piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q1/5

$g(x) = -1$ for $x < 0$, $g(x) = 2$ for $0 \le x < 3$, and $g(x) = x - 1$ for $x \ge 3$.

Graph g. Use open and closed dots at the breaks.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q2/5

1622

QUESTION ID

S75-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A horizontal ray at y = −1 for x < 0 with an open dot at (0, −1); a segment at y = 2 from a closed dot at (0, 2) to an open dot at (3, 2); a ray from a closed dot at (3, 2) rising with slope 1

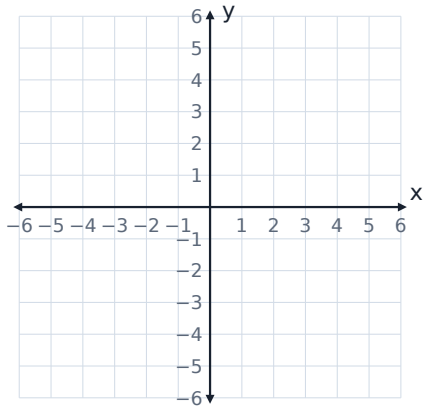
GRADING NOTE Any correct graph with the correct open and closed dots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph the greatest integer function y = ⌊x⌋ (the greatest integer less than or equal to x) for −2 ≤ x < 3.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q3/5

1623

QUESTION ID

S75-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Horizontal steps of length 1: y = −2 for −2 ≤ x < −1, y = −1 for −1 ≤ x < 0, y = 0 for 0 ≤ x < 1, y = 1 for 1 ≤ x < 2, y = 2 for 2 ≤ x < 3; each step has a closed left end and an open right end

GRADING NOTE Any correct graph with the closed and open ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A parking garage charges $3 for each hour or part of an hour.

What is the charge for 2.25 hours?

A. $6

B. $6.75

C. $7.50

D. $9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q4/5

1624

QUESTION ID

S75-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $9

GRADING NOTE 2.25 hours is 3 started hours: 3 × $3 = $9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q4/5

True or false?

For the greatest integer function, ⌊2.7⌋ = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q5/5

1625

QUESTION ID

S75-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ⌊2.7⌋ is the greatest integer less than or equal to 2.7, which is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q5/5

SET 75 • HSF-IF.C.7.b

BACKWARD 1

Evaluate piecewise-defined functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x - 1$ for $x < 3$, and $f(x) = 8$ for $x \geq 3$.

Find $f(3)$ and $f(1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q1/5

1627

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

f(3) = 8 and f(1) = 1

GRADING NOTE Form required: both values. x = 3 uses the second piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = |x|$ for $x \le 0$, and $g(x) = x^2$ for $x > 0$.

Find $g(-4)$ and $g(4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q2/5

1628

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$g(-4) = 4$ and $g(4) = 16$

GRADING NOTE Form required: both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(x) = 3$ for $x < 1$, and $h(x) = 2x$ for $x \geq 1$. What is $h(1)$?

A. 2

B. 5

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q3/5

1629

QUESTION ID

S75-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE $x = 1$ uses the second piece: $2(1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q3/5

True or false?

$p(x) = 5$ for $x \leq 2$, and $p(x) = x$ for $x > 2$. Then $p(2) = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q4/5

1630

QUESTION ID

S75-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = 2$ uses the first piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\lfloor -1.5 \rfloor = -1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q5/5

1631

QUESTION ID

S75-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The greatest integer less than or equal to −1.5 is −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.b

BACKWARD 2

Order numbers and describe intervals on a number line • Grade 6 • The Number System • 6.NS.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Is −3 to the left or to the right of −1 on a number line? Which number is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q1/5

1633

QUESTION ID

S75-B2-Q1

TYPE

Short answer

CORRECT ANSWER

To the left; −1 is greater.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an inequality for "x is at least 0 and less than 3".

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q2/5

1634

QUESTION ID

S75-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \leq x < 3$

GRADING NOTE Form required: a compound inequality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which statement is true?

A. $-4 > -2$

B. $-4 < -2$

C. $0 < -1$

D. $-2 < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q3/5

1635

QUESTION ID

S75-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $-4 < -2$

GRADING NOTE -4 is to the left of -2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

−5 is less than −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q4/5

1636

QUESTION ID

S75-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −5 is to the left of −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q4/5

True or false?

On a number line, 2 is to the left of −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q5/5

1637

QUESTION ID

S75-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2 is to the right of −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q5/5

FORWARD 1

Graph the probability distribution of a discrete random variable • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: A probability distribution assigns a value to each separate outcome, like a function defined piece by piece, and its bar graph is read the same way.

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped twice, and X is the number of heads. Give P(X = 0), P(X = 1), and P(X = 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q1/5 • NEAREST RELATED

1639

QUESTION ID
S75-F1-Q1

TYPE
Short answer

CORRECT ANSWER
1/4, 1/2, 1/4

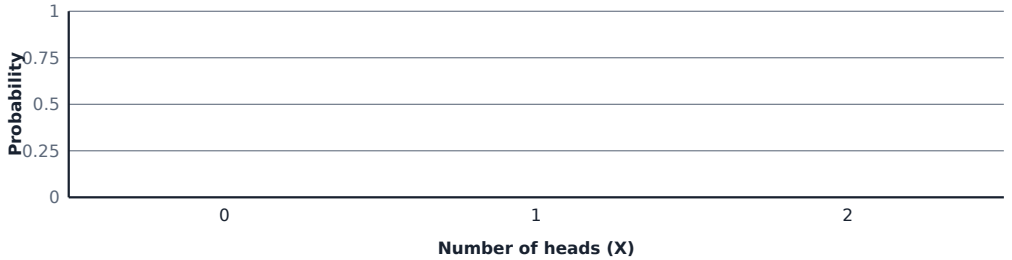
GRADING NOTE Form required: all three. The outcomes HH, HT, TH, TT are equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped twice, and X is the number of heads. Graph the probability distribution of X as a bar graph.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q2/5 • NEAREST RELATED

1640

QUESTION ID
S75-F1-Q2

TYPE
Drawing / plotting

CORRECT ANSWER
Bars of height 0.25 at X = 0, 0.5 at X = 1, and 0.25 at X = 2

GRADING NOTE Any correct bar graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

X is the result of one roll of a fair number cube numbered 1 to 6. What is P(X ≥ 5)?

A. $\frac{5}{6}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q3/5 • NEAREST RELATED

1641

QUESTION ID

S75-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/3

GRADING NOTE 5 and 6: 2 of 6 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

The probabilities of a probability distribution add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q4/5 • NEAREST RELATED

1642

QUESTION ID
S75-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The outcomes cover every possibility.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

For X = the number of heads in 2 fair coin flips, $P(X = 1) = \frac{1}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q5/5 • NEAREST RELATED

1643

QUESTION ID
S75-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE HT and TH are 2 of 4 equally likely outcomes, so $P(X = 1) = \frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 75 • HSF-IF.C.7.b

FORWARD 2

Read inverse values of piecewise and step functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$. This f is one-to-one. Find $f^{-1}(9)$ and $f^{-1}(-4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q1/5

1645

QUESTION ID
S75-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(9) = 3$ and $f^{-1}(-4) = -2$

GRADING NOTE Form required: both values. $3^2 = 9$ with $3 \geq 0$, and $2(-2) = -4$ with $-2 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q1/5

$g(x) = x + 1$ for $x < 2$, and $g(x) = 2x - 1$ for $x \geq 2$. This g is one-to-one. Find $g^{-1}(7)$.

$g^{-1}(7) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q2/5

1646

QUESTION ID

S75-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $x + 1 = 7$ gives 6, which is not less than 2; $2x - 1 = 7$ gives 4, which is at least 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q2/5

Why does the greatest integer function $y = \lfloor x \rfloor$ have no inverse function?

A. Many inputs share each output, such as $\lfloor 2.1 \rfloor = \lfloor 2.9 \rfloor = 2$.

B. It is a linear function.

C. It is not defined for negative x.

D. Its outputs are not integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q3/5

1647

QUESTION ID

S75-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Many inputs share each output, such as $\lfloor 2.1 \rfloor = \lfloor 2.9 \rfloor = 2$.

GRADING NOTE A function needs to be one-to-one to have an inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q3/5

True or false?

A piecewise function whose pieces are increasing and whose output ranges do not overlap can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q4/5

1648

QUESTION ID

S75-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then every output comes from exactly one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q4/5

True or false?

If f is one-to-one and $f(3) = 5$, then $f^{-1}(3) = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q5/5

1649

QUESTION ID
S75-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $f^{-1}(5) = 3$; $f^{-1}(3)$ need not be 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions and show intercepts and end behavior

MAIN	Algebra I main question	**HSF-IF.C.7.e**
BACKWARD 1	Evaluate integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate exponential functions	**HSF-IF.A.2**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Understand logarithms as inverses of exponential functions	**HSF-BF.B.5**

SET 44 • HSF-IF.C.7.e

MAIN

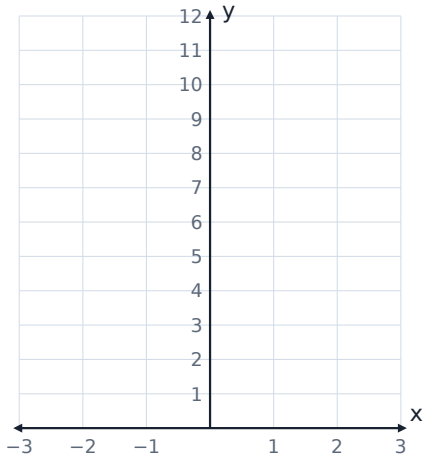
Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = 3 \cdot 2^x$ for $-2 \le x \le 2$.

Give its y-intercept and its horizontal asymptote.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q1/5

1652

QUESTION ID

S44-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A curve through (−2, 0.75), (−1, 1.5), (0, 3), (1, 6), and (2, 12), rising more steeply to the right

Written: y-intercept (0, 3); asymptote y = 0

GRADING NOTE Form required: both. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q1/5

$g(x) = 50(0.8)^x$.

Is this growth or decay? Give its y-intercept, and say what happens to g(x) as x gets large.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q2/5

1653

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

Decay; y-intercept 50; g(x) gets closer and closer to 0.

GRADING NOTE Form required: all three parts. 0.8 < 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q2/5

$h(x) = 2^x - 4$.

Find the y-intercept, the x-intercept, and the horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q3/5

1654

QUESTION ID
S44-M-Q3

TYPE
Short answer

CORRECT ANSWER
y-intercept (0, −3); x-intercept (2, 0); asymptote y = −4

GRADING NOTE Form required: all three. $2^0 - 4 = -3$, and $2^x = 4$ at $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function's graph rises from left to right and passes through (0, 5)?

A. $y = 1.2(5)^x$

B. $y = 0.8(5)^x$

C. $y = 5(0.8)^x$

D. $y = 5(1.2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q4/5

1655

QUESTION ID

S44-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 5(1.2)^x$

GRADING NOTE A base greater than 1 gives growth, and the coefficient is the y-intercept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of $y = 4(0.5)^x$ crosses the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q5/5

1656

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4(0.5)^x$ is always positive; the x-axis is an asymptote.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q5/5

SET 44 • HSF-IF.C.7.e

BACKWARD 1

Evaluate integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2^{-2}.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

1658

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form, such as 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $3 \cdot 2^{-1}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

1659

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE Any equivalent form, such as 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which number equals $(0.5)^{-1}$?

A. -0.5

B. 0.5

C. 2

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

1660

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $1 \div 0.5 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

$5^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

1661

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any nonzero number to the 0 power is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4^{-1} = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

1662

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^{-1} = 1 \div 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

BACKWARD 2

Evaluate exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 50(0.8)^x$.

Find f(2).

f(2) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q1/5

1664

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form. 50(0.64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 2^x - 4$.

Find g(3).

g(3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q2/5

1665

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. 8 − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q2/5

$h(x) = 3 \cdot 2^x$.

Find $h(-1)$.

A. -1.5

B. 6

C. -6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q3/5

1666

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1.5

GRADING NOTE $3 \cdot \frac{1}{2} = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q3/5

True or false?

If $f(x) = 10^x$, then $f(3) = 1{,}000$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q4/5

1667

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^3 = 1{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q4/5

True or false?

If $f(x) = 3^x$, then $f(2) = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q5/5

1668

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3 \cdot 2^x = 48$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q1/5

1670

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $2^x = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $50(0.8)^x = 20$. Round to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q2/5

1671

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.1

GRADING NOTE Form required: rounded to the nearest tenth. $x = \log 0.4 \div \log 0.8 \approx 4.11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2^x = 12$. (log means base 10.)

A. x = log 6

B. x = log 12 ÷ log 2 ≈ 3.58

C. x = 3.5

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q3/5

1672

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = log 12 ÷ log 2 ≈ 3.58

GRADING NOTE Take the logarithm of both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

log 1 = 0 (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q4/5

1673

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

log 100 = 2 (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q5/5

1674

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^2 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q5**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 2

Understand logarithms as inverses of exponential functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $f(x) = 2^x$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q1/5

1676

QUESTION ID
S44-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$\log_2 x$

GRADING NOTE Any equivalent form, such as log x ÷ log 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\log_3 81 = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q2/5

1677

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$3^4 = 81$, and $\log_3 81$ is the exponent on 3 that gives 81.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q2/5

The graph of $y = \log_2 x$ is the reflection of the graph of $y = 2^x$ across which line?

A. $x = 0$

B. $y = 0$

C. $y = x$

D. $y = -x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q3/5

1678

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = x

GRADING NOTE Inverse functions reflect across y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q3/5

True or false?

The domain of y = log x is x > 0. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q4/5

1679

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the range of $y = 10^x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q4/5

True or false?

$\log_2(2^5) = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q5/5

1680

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log_2(2^5) = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSF-IF.C.8.a • INTERPRETING FUNCTIONS

Factor and complete the square to show zeros, extreme values, and symmetry, and interpret them

MAIN	Algebra I main question	**HSF-IF.C.8.a**
BACKWARD 1	Complete the square	**HSA-SSE.B.3.b**
BACKWARD 2	Use the greatest common factor	**6.NS.B.4**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Complete the square to find the center of a hyperbola	**HSG-GPE.A.3**

SET 45 • HSF-IF.C.8.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

$f(x) = x^2 - 8x + 12$.

Write f in factored form and in vertex form. Give the zeros, the vertex, and the axis of symmetry.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q1/5

1683

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x − 6) and (x − 4)² − 4; zeros 2 and 6; vertex (4, −4); axis x = 4

GRADING NOTE Form required: both forms and all three features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A company's profit is $P(x) = -2x^2 + 120x - 1{,}000$ dollars when it charges x dollars.

Complete the square to find the price that gives the most profit, and the greatest profit.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q2/5

1684

QUESTION ID

S45-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$P(x) = -2(x - 30)^2 + 800$; a price of $30 gives the greatest profit, $800.

GRADING NOTE REQUIRED WORK: The completed square shown:
$-2(x^2 - 60x) - 1{,}000 = -2(x^2 - 60x + 900) + 1{,}800 - 1{,}000 = -2(x - 30)^2 + 800$.

Form required: the completed square with its steps, the price, and the profit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A rock's height is $h(t) = -16t^2 + 32t + 48$ feet after t seconds.

Factor to find when it hits the ground.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q3/5

1685

QUESTION ID
S45-M-Q3

TYPE
Short answer

CORRECT ANSWER
$h(t) = -16(t - 3)(t + 1)$; at t = 3 seconds

GRADING NOTE Form required: the factored form and the time. $t = -1$ is not in the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which form of $f(x) = x^2 + 6x + 5$ shows the vertex directly?

A. $(x + 1)(x + 5)$

B. $x^2 + 6x + 5$

C. $(x + 3)^2 - 4$

D. $x(x + 6) + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q4/5

1686

QUESTION ID

S45-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(x + 3)^2 - 4$

GRADING NOTE Vertex form $a(x - h)^2 + k$ shows the vertex $(-3, -4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q4/5

True or false?

The axis of symmetry of $y = (x - 1)(x - 7)$ is $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q5/5

1687

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The axis is halfway between the zeros 1 and 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q5/5

SET 45 • HSF-IF.C.8.a

BACKWARD 1

Complete the square • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square.

$x^2 - 8x + 12$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q1/5

1689

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 4)^2 - 4$

GRADING NOTE Form required: $(x - h)^2 + k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square.

$x^2 + 3x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q2/5

1690

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1.5)^2 - 2.25$

GRADING NOTE Form required: $(x - h)^2 + k$, with fractions or decimals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q2/5

1690

ANSWER KEY Reduced copy of the question page, with its answer

Which expression equals $-(x^2 - 6x)$?

A. $-(x - 3)^2 - 9$

B. $-(x - 3)^2 + 9$

C. $(x - 3)^2 - 9$

D. $-(x + 3)^2 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q3/5

1691

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $-(x - 3)^2 + 9$

GRADING NOTE $x^2 - 6x = (x - 3)^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 2x + 10 = (x + 1)^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q4/5

1692

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + 1)^2 = x^2 + 2x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 5)^2 - 3$ has its minimum at $x = -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q5/5

1693

QUESTION ID
S45-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its minimum is at $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.C.8.a

BACKWARD 2

Use the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Use the greatest common factor to write 16 + 40 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q1/5

1695

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8(2 + 5)

GRADING NOTE Form required: the GCF times a sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the greatest common factor of 32 and 48.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q2/5

1696

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q2/5

Which expression equals 36 + 27 and uses the greatest common factor?

A. 9(4 + 3)

B. 3(12 + 6)

C. 6(6 + 27)

D. 9(4 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q3/5

1697

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9(4 + 3)

GRADING NOTE $9 \cdot 4 = 36$ and $9 \cdot 3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q3/5

True or false?

The greatest common factor of 12 and 18 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q4/5

1698

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 6 divides both, and no larger number does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$20 + 50 = 5(4 + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q5/5

1699

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5(4 + 5) = 45$. $20 + 50 = 10(2 + 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.C.8.a

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

ANSWER KEY Reduced copy of the question page, with its answer

Show that $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ by expanding both sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q1/5

1701

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Left: $x^4 + 2x^2y^2 + y^4$. Right: $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The two sides are equal.

GRADING NOTE Form required: both sides expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $x = 2$ and $y = 1$ in $x^2 - y^2$, $2xy$, and $x^2 + y^2$ to make a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q2/5

1702

QUESTION ID
S45-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3, 4, 5

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q2/5

Which equation is an identity (true for all numbers a and b)?

A. $a^3 + b^3 = (a + b)^3$

B. $a^3 - b^3 = (a - b)^3$

C. $a^2 + b^2 = (a + b)^2$

D. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q3/5

1703

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D.

$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

GRADING NOTE Expanding $(a - b)(a^2 + ab + b^2)$ gives $a^3 - b^3$. Each other choice fails for $a = 2$, $b = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q3/5

True or false?

$(a + b)^2 - (a - b)^2 = 4ab$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q4/5

1704

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q4/5

True or false?

$(x + 1)^3 = x^3 + 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q5/5

1705

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSF-IF.C.8.a

FORWARD 2

Complete the square to find the center of a hyperbola • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to find the center of the hyperbola $x^2 - 4y^2 - 6x - 8y - 11 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q1/5

1707

QUESTION ID
S45-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(3, −1)

GRADING NOTE Any equivalent form. $(x - 3)^2 - 4(y + 1)^2 = 11 + 9 - 4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q1/5

The hyperbola $x^2 \div 16 - y^2 \div 9 = 1$ has foci (5, 0) and (−5, 0).

Show that (5, 2.25) is on the hyperbola and that the absolute difference of its distances to the foci is 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q2/5

1708

QUESTION ID

S45-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

It satisfies the equation, and $|10.25 - 2.25| = 8$.

GRADING NOTE

REQUIRED WORK: $25 \div 16 - 5.0625 \div 9 = 1.5625 - 0.5625 = 1$ shown; the distances 2.25 and $\sqrt{(100 + 5.0625)} = 10.25$ computed and subtracted.

Form required: the substitution and both distances. $8 = 2a$ with $a = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation is a hyperbola?

A. $y = x^2 \div 4$

B. $x^2 \div 4 + y^2 \div 9 = 1$

C. $x^2 \div 4 - y^2 \div 9 = 1$

D. $x^2 + y^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q3/5

1709

QUESTION ID
S45-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $x^2 \div 4 - y^2 \div 9 = 1$

GRADING NOTE A hyperbola has a difference of squared terms; $x^2 \div 4 + y^2 \div 9 = 1$ is an ellipse, $y = x^2 \div 4$ a parabola, and $x^2 + y^2 = 9$ a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q3/5

True or false?

For a hyperbola, the absolute value of the difference of the distances from any of its points to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q4/5

1710

QUESTION ID
S45-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the definition of a hyperbola. The signed difference has opposite signs on the two branches, so the absolute value is used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q4/5

True or false?

The hyperbola $x^2 - y^2 = 1$ passes through (0, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q5/5

1711

QUESTION ID
S45-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $0 - 1 = -1$, not 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Use the properties of exponents to interpret growth and decay rates

MAIN	Algebra I main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth and decay from an exponential function	**HSF-IF.C.7.e**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Sum a geometric series in a savings context	**HSA-SSE.B.4**
FORWARD 2	Find times in exponential models with logarithms	**HSF-LE.A.4**

SET 46 • HSF-IF.C.8.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

A savings balance is $y = 800(1.035)^t$ dollars after t years.

Is this growth or decay, and what is the percent rate of change per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q1/5

1714

QUESTION ID
S46-M-Q1

TYPE
Short answer

CORRECT ANSWER
Growth, at 3.5% per year

GRADING NOTE Form required: growth or decay and the percent. 1.035 = 1 + 0.035.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A loan balance grows by 1.5% each month:

$B = 2{,}000(1.015)^{12t}$, where t is in years.

Rewrite it to find the yearly growth factor (to four decimal places) and the yearly percent rate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q2/5

1715

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

$B = 2{,}000(1.015^{12})^{t} \approx 2{,}000(1.1956)^{t}$; about 19.56% per year

GRADING NOTE Form required: the yearly factor and the percent. $(1.015)^{12t} = ((1.015)^{12})^{t}$, and $1.015^{12} \approx 1.19562$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q2/5

The value of a machine is $V = 45{,}000(0.88)^t$ dollars after t years.

What is the yearly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q3/5

1716

QUESTION ID

S46-M-Q3

TYPE

Short answer

CORRECT ANSWER

12% per year

GRADING NOTE Any equivalent form. $0.88 = 1 - 0.12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A forest's area is $A = 600(1.3)^{t/20}$ hectares after t years.

By what percent does A grow every 20 years?

A. 1.3%

B. 13%

C. 3%

D. 30%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q4/5

1717

QUESTION ID

S46-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 30%

GRADING NOTE When t increases by 20, the exponent increases by 1, so A is multiplied by 1.3: a 30% increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 100(0.5)^{t/8}$ halves every 8 units of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q5/5

1718

QUESTION ID
S46-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each increase of 8 in t multiplies y by 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q5/5

SET 46 • HSF-IF.C.8.b

BACKWARD 1

Read growth and decay from an exponential function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = 30(1.5)^x$ increasing or decreasing? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q1/5

1720

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Increasing, because the base 1.5 is greater than 1.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the y-intercept of $y = 7(0.9)^x$?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q2/5

1721

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form, such as (0, 7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q2/5

Which function is decreasing?

A. $y = 2(1.6)^x$

B. $y = 0.6(1.2)^x$

C. $y = 2(0.6)^x$

D. $y = 0.6(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q3/5

1722

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2(0.6)^x$

GRADING NOTE $y = 2(0.6)^x$ is the only one whose base, 0.6, is between 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q3/5

True or false?

The graph of $y = 3^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q4/5

1723

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3^x approaches 0 as x decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q4/5

True or false?

$y = 5(1.1)^x$ has y-intercept 1.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q5/5

1724

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0, y = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.8.b

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A $40 shirt is marked up 15%.

What is the new price?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q1/5

1726

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$46

GRADING NOTE Any equivalent form. 40 × 1.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A town grows from 2,000 to 2,300 people.

What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q2/5

1727

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

15%

GRADING NOTE Any equivalent form. 300 ÷ 2,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A price drops from $80 to $60.

What is the percent decrease?

A. 25%

B. 20%

C. 33%

D. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q3/5

1728

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 25%

GRADING NOTE 20 ÷ 80 = 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying by 1.06 increases a number by 6%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q4/5

1729

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1.06 = 1 + 0.06.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying by 0.85 decreases a number by 85%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q5/5

1730

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases the number by 15%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSF-IF.C.8.b

FORWARD 1

Sum a geometric series in a savings context • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

You save $100 the first month and 2% more each month after that.

Write the total for 12 months as a sum formula, and find it to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q1/5

1732

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$100(1.02^{12} - 1) \div 0.02 \approx \$1{,}341$

GRADING NOTE Form required: the formula and the rounded total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $4 + 4(1.5) + 4(1.5)^2 + 4(1.5)^3$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q2/5

1733

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

32.5

GRADING NOTE Any equivalent form. 4 + 6 + 9 + 13.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression is the sum of the first n terms of $a + ar + ar^2 + \ldots$ $(r \neq 1)$?

A. $a \div (1 - r)$

B. $a(1 - r) \div (1 - r^n)$

C. $a(1 - r^n) \div (1 - r)$

D. nar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q3/5

1734

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $a(1 - r^n) \div (1 - r)$

GRADING NOTE The finite geometric series formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1 + 0.5 + 0.25 + 0.125 = 1.875$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q4/5

1735

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add the terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q4/5

True or false?

The sum of 10 terms of $3 + 3 + 3 + \ldots$ is 3^{10}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q5/5

1736

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10 \cdot 3 = 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSF-IF.C.8.b

FORWARD 2

Find times in exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$y = 100(0.5)^{t/8}$.

Find t when y = 10. Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q1/5

1738

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

26.6

GRADING NOTE Form required: rounded to the nearest tenth. t = 8 · log 0.1 ÷ log 0.5 ≈ 26.58.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the continuous rate r for which $e^{10r} = 2$. Round to the nearest thousandth.

r ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q2/5

1739

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.069

GRADING NOTE Form required: rounded to the nearest thousandth. r = ln 2 ÷ 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

To the nearest tenth, how long does $y = (1.05)^t$ take to double?

A. 20

B. 7.2

C. 10.5

D. 14.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q3/5

1740

QUESTION ID
S46-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 14.2

GRADING NOTE $t = \log 2 \div \log 1.05 \approx 14.21$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q3/5

True or false?

If $y = 2^{t/3}$, then y doubles every 3 units of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S46-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q4/5

1741

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Adding 3 to t adds 1 to the exponent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q4/5

True or false?

$\log(2^{10}) = 20$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q5/5

1742

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(2^{10}) = 10 \log 2 \approx 3.01$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q5**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q5/5

SET 47 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of two functions represented in different ways

MAIN	Algebra I main question	**HSF-IF.C.9**
BACKWARD 1	Interpret key features in context	**HSF-IF.B.4**
BACKWARD 2	Compute average rates of change	**HSF-IF.B.6**
FORWARD 1	Compare periodic functions given in different forms	**HSF-TF.B.5**
FORWARD 2	Compare normal distributions by mean and standard deviation —…	**HSS-ID.A.4**

SET 47 • HSF-IF.C.9

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 3x + 2. The table shows g.

Which function has the greater value at x = 3? Which has the greater average rate of change from x = 0 to x = 3?

x	0	1	2	3
g(x)	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q1/5

1745

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

g at x = 3 (16 compared with 11); g has the greater rate (5 compared with 3).

GRADING NOTE Form required: both comparisons with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q1/5

Function A is $y = -2(x - 3)^2 + 10$. Function B is a parabola that opens down with vertex (1, 12).

Which has the greater maximum value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q2/5

1746

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

B: its maximum is 12, and A's is 10.

GRADING NOTE Form required: the function and both maxima.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Function f has y-intercept 4 and increases by 5 for each 1-unit increase in x. g(x) = 5x − 1.

Compare their slopes and y-intercepts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q3/5

1747

QUESTION ID

S47-M-Q3

TYPE

Short answer

CORRECT ANSWER

Same slope, 5; f has the greater y-intercept (4 compared with −1), so their graphs are parallel lines.

GRADING NOTE Form required: both comparisons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$P(x) = 1.08^x$ grows 8% per year. The table shows Q.

Which grows faster?

x	0	1	2
Q(x)	1	1.1	1.21

A. P: its graph is a line.

B. Q: it grows 10% per year.

C. P: it grows 8% per year and Q grows 1%.

D. They grow at the same rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q4/5

1748

QUESTION ID

S47-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Q: it grows 10% per year.

GRADING NOTE Q is multiplied by 1.1 each year: 1, 1.1, 1.21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f(x) = x^2$, and g is a parabola opening up with vertex (0, −4). The two functions have the same minimum value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q5/5

1749

QUESTION ID
S47-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The minimum of f is 0 and the minimum of g is −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 1

Interpret key features in context • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

T(h) is the temperature h hours after midnight, and its minimum is T(6) = 4.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q1/5

1751

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The lowest temperature, 4 degrees, happened 6 hours after midnight (6 a.m.).

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph of savings S(m) is horizontal from m = 3 to m = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q2/5

1752

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The savings stayed the same from month 3 to month 5.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A function's graph is an unbroken curve. It is positive for $0 < x < 8$ and negative for $x > 8$.

What is true at $x = 8$?

A. The output is greatest.

B. The output is 8.

C. The output is 0.

D. The function is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q3/5

1753

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The output is 0.

GRADING NOTE An unbroken curve that changes sign passes through the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A y-intercept gives the starting value of a quantity that changes over time.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q4/5

1754

QUESTION ID
S47-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It is the output at time 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

An x-intercept tells the starting value of a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q5/5

1755

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An x-intercept is where the output is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 2

Compute average rates of change • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2^x$.

Find the average rate of change from $x = 1$ to $x = 3$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q1/5

1757

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. (8 − 2) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A savings account had $100 at week 0 and $180 at week 4.

What is the average rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q2/5

1758

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$20 per week

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = x^2 - 1$.

What is the average rate of change from $x = -1$ to $x = 2$?

A. 1

B. -1

C. 3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q3/5

1759

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1

GRADING NOTE $(3 - 0) \div 3 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q3/5

True or false?

The average rate of change of $y = 5$ is 0 on every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q4/5

1760

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The output never changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q4/5

1760

True or false?

The average rate of change of $y = x^2$ from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q5/5

1761

QUESTION ID
S47-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(4 - 0) \div 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.9

FORWARD 1

Compare periodic functions given in different forms • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 3sin(2x). Function g has amplitude 4 and period π.

Which has the greater amplitude? Do they have the same period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q1/5

1763

QUESTION ID
S47-F1-Q1

TYPE
Short answer

CORRECT ANSWER
g has the greater amplitude (4 compared with 3); yes, both periods are π.

GRADING NOTE Form required: both parts. The period of f is 2π ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tide A rises to 12 ft and falls to 2 ft. Tide B is $h(t) = 6 + 4\sin t$.

Which has the greater amplitude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q2/5

1764

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Tide A: 5 ft compared with 4 ft.

GRADING NOTE Form required: the tide and both amplitudes. (12 − 2) ÷ 2 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q2/5

What is the period of y = sin(4x)?

A. π ÷ 2

B. π ÷ 4

C. 2π

D. 4π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q3/5

1765

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. π ÷ 2

GRADING NOTE 2π ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q3/5

True or false?

y = 2cos x and y = 2sin x have the same amplitude and the same period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q4/5

1766

QUESTION ID

S47-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both have amplitude 2 and period 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q4/5

True or false?

y = 5sin x has a maximum value of 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q5/5

1767

QUESTION ID
S47-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its maximum value is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSF-IF.C.9

FORWARD 2

Compare normal distributions by mean and standard deviation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Comparing two functions by their key features leads to comparing two normal curves by their center and spread.

ANSWER KEY Reduced copy of the question page, with its answer

Test A scores are normal with mean 70 and standard deviation 5. Test B scores are normal with mean 70 and standard deviation 10.

Which test's scores are more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

1769

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Test B: its standard deviation is larger.

GRADING NOTE Form required: the test and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Heights are normal with mean 64 in and standard deviation 3 in.

About what percent are between 61 in and 67 in?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

1770

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

About 68%

GRADING NOTE Any equivalent form. That interval is within 1 standard deviation of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Scores are normal with mean 100 and standard deviation 15.

About what percent of scores are above 130?

A. About 16%

B. About 34%

C. About 2.5%

D. About 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

1771

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. About 2.5%

GRADING NOTE 130 is 2 standard deviations above the mean; about 95% lie within 2, so about 2.5% lie above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A normal distribution is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

1772

QUESTION ID

S47-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its curve is a symmetric bell shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

In a normal distribution, about 95% of the data lie within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

1773

QUESTION ID
S47-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE About 68% lie within 1; about 95% lie within 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

Write an explicit expression or a recursive process for a function from a context

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Evaluate a sequence from a recursive or explicit rule	**HSF-IF.A.3**
BACKWARD 2	Write equations in two or more variables	**HSA-CED.A.2**
FORWARD 1	Build a composite function from a context	**HSF-BF.A.1.c**
FORWARD 2	Find and interpret the inverse of a model	**HSF-BF.B.4.b**

SET 48 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A phone plan has a $60 activation fee and costs $35 per month.

Write an explicit expression for the total cost C(n) after n months, and a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q1/5

1776

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

C(n) = 60 + 35n; C(0) = 60 and C(n) = C(n − 1) + 35 for n ≥ 1

GRADING NOTE Form required: both the explicit expression and the recursive rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A tree is 4 ft tall and grows 1.5 ft per year.

Write a function for its height h(t) after t years.

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q2/5

1777

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 + 1.5t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball is dropped from 10 m. Each bounce reaches 60% of the height of the bounce before.

Write a function for the height h(n) of the nth bounce.

h(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q3/5

1778

QUESTION ID

S48-M-Q3

TYPE

Short answer

CORRECT ANSWER

$10(0.6)^n$

GRADING NOTE Any equivalent form. The first bounce reaches 10(0.6) = 6 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows n = 1, 2, 3, 4 and a(n) = 7, 11, 15, 19.

Which recursive process fits?

A. a(1) = 4, a(n) = a(n − 1) + 7

B. a(1) = 7, a(n) = 4a(n − 1)

C. a(1) = 7, a(n) = a(n − 1) + 4

D. a(n) = 7n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q4/5

1779

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a(1) = 7, a(n) = a(n − 1) + 4

GRADING NOTE Start at 7 and add 4 each time. (7n + 4 gives 11 for n = 1.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The steps "start at 500 and multiply by 1.03 each year" give f(t) = 500 + 1.03t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q5/5

1780

QUESTION ID

S48-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Repeated multiplication gives $f(t) = 500(1.03)^t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 1

Evaluate a sequence from a recursive or explicit rule • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 2$ and $f(n) = 3f(n - 1)$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q1/5

1782

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

54

GRADING NOTE Any equivalent form. 2, 6, 18, 54.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 10 - 2n$ for $n \geq 1$.

List the first four terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q2/5

1783

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8, 6, 4, 2

GRADING NOTE Form required: the four terms in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 5$ and $f(n) = f(n - 1) - 3$.

What is $f(3)$?

A. -1

B. 11

C. 2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q3/5

1784

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −1

GRADING NOTE 5, 2, −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q3/5

True or false?

The terms f(1), f(2), f(3) of a sequence are its outputs for the inputs 1, 2, 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q4/5

1785

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A sequence is a function on the integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f(n) = n^2$ gives the terms 1, 2, 3, 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q5/5

1786

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives 1, 4, 9, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 2

Write equations in two or more variables • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A car rental costs $45 per day plus $0.20 per mile.

Write an equation for the cost C for d days and m miles.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q1/5

1788

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

C = 45d + 0.20m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation for the perimeter P of a rectangle with length l and width w.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q2/5

1789

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

P = 2l + 2w

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A $900 laptop loses $150 of value each year.

Which equation gives its value V after t years?

A. $V = 900 - 150t$

B. $V = 150 - 900t$

C. $V = 900(150)^t$

D. $V = 900 + 150t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q3/5

1790

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. V = 900 − 150t

GRADING NOTE Start at 900 and subtract 150 per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q3/5

True or false?

$A = s^2$ gives the area of a square with side s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q4/5

1791

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Side times side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 12x gives the number of feet in x inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q5/5

1792

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE y = 12x gives the number of inches in x feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 1

Build a composite function from a context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

An oil spill's radius is $r(t) = 2t$ meters after t hours, and the area of a circle is $A(r) = \pi r^2$.

Write $A(r(t))$ and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q1/5

1794

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$A(r(t)) = 4\pi t^2$: the area of the spill after t hours.

GRADING NOTE Any equivalent form; the meaning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A store takes 20% off and then $5 off: d(x) = 0.8x and c(x) = x − 5.

Find c(d(50)).

c(d(50)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q2/5

1795

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

35

GRADING NOTE Any equivalent form. d(50) = 40, and c(40) = 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

With d(x) = 0.8x and c(x) = x − 5, which expression is d(c(x))?

A. 0.8x − 4

B. 0.8x + 4

C. 0.8x − 5

D. x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q3/5

1796

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.8x − 4

GRADING NOTE 0.8(x − 5) = 0.8x − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q3/5

True or false?

A store takes 20% off and then $5 off: d(x) = 0.8x and c(x) = x − 5. For these functions, c(d(50)) and d(c(50)) are different.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q4/5

1797

QUESTION ID
S48-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE c(d(50)) = 35 and d(c(50)) = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q4/5

True or false?

If f(x) = 3x and g(x) = x + 2, then f(g(x)) = 3x + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q5/5

1798

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(g(x)) = 3(x + 2) = 3x + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 2

Find and interpret the inverse of a model • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

C(n) = 60 + 35n is the total cost after n months.

Find the inverse and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q1/5

1800

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

n = (C − 60) ÷ 35: the number of months for a total cost of C dollars.

GRADING NOTE Any equivalent form; the meaning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that f(x) = 5x − 2 and g(x) = (x + 2) ÷ 5 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q2/5

1801

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f(g(x)) = (x + 2) − 2 = x and g(f(x)) = (5x − 2 + 2) ÷ 5 = x.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q2/5

F = 1.8C + 32 converts Celsius to Fahrenheit.

Which formula is its inverse?

A. C = F ÷ 1.8 − 32

B. C = 1.8F + 32

C. C = (F + 32) ÷ 1.8

D. C = (F − 32) ÷ 1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q3/5

1802

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. C = (F − 32) ÷ 1.8

GRADING NOTE Subtract 32, then divide by 1.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q3/5

True or false?

If f is invertible and $f(2) = 9$, then $f^{-1}(9) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q4/5

1803

QUESTION ID

S48-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The inverse reverses each input-output pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = 2x$ is $f^{-1}(x) = -2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q5/5

1804

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f^{-1}(x) = x \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q5/5

Build quadratic functions that model situations

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Write equations in two variables for areas and perimeters	**HSA-CED.A.2**
BACKWARD 2	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
FORWARD 1	Use quadratic models to make the best design choice	**HSG-MG.A.3**
FORWARD 2	Compose functions to build quadratic models	**HSF-BF.A.1.c**

SET 80 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A farmer has 100 m of fence for three sides of a rectangular pen; a barn wall forms the fourth side. Each side perpendicular to the wall is x m long.

Write the area A(x) of the pen.

A(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q1/5

1807

QUESTION ID

S80-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x(100 - 2x)$

GRADING NOTE Any equivalent form, such as $100x - 2x^2$. The side parallel to the wall uses the remaining $100 - 2x$ meters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A theater sells 300 tickets at $20 each. For each $1 increase in price, it sells 10 fewer tickets.

Write the revenue R(n) after n price increases of $1.

R(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q2/5

1808

QUESTION ID

S80-M-Q2

TYPE

Short answer

CORRECT ANSWER

(20 + n)(300 − 10n)

GRADING NOTE Any equivalent form, such as $-10n^2 + 100n + 6{,}000$. Revenue = price × tickets sold.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A stone is thrown upward at 20 m/s from the top of a 5-m wall. Using $h(t) = -4.9t^2 + v_0t + h_0$, where v_0 is the starting upward speed and h_0 is the starting height, write the stone's height h(t).

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q3/5

1809

QUESTION ID

S80-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-4.9t^2 + 20t + 5$

GRADING NOTE Any equivalent form. $v_0 = 20$ and $h_0 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table has x = 0, 1, 2, 3 and y = 3, 5, 11, 21.

Which function fits the table?

A. $y = 3(2)^x$

B. $y = 2x + 3$

C. $y = 2x^2 + 3$

D. $y = x^2 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q4/5

1810

QUESTION ID

S80-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2x^2 + 3$

GRADING NOTE The second differences are all 4, so the x^2 coefficient is 2; y(0) = 3. Check: 2(9) + 3 = 21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If each of n people shakes hands once with every other person, the number of handshakes is $h(n) = n(n - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q5/5

1811

QUESTION ID

S80-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each handshake is counted twice in $n(n - 1)$, so $h(n) = n(n - 1) \div 2$ (still a quadratic function).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.A.1.a

BACKWARD 1

Write equations in two variables for areas and perimeters • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has length 2w and width w. Write an equation for its perimeter P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q1/5

1813

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6w

GRADING NOTE Any equivalent form, such as 2(2w) + 2w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has base b and height 2b. Write an equation for its area A.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q2/5

1814

QUESTION ID

S80-B1-Q2

TYPE

Short answer

CORRECT ANSWER

b^2

GRADING NOTE Any equivalent form, such as $\frac{1}{2}(b)(2b)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q2**

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q2/5

A rectangle has perimeter 40 and width w. Which equation gives its area A?

A. $A = w^2 + 20$

B. $A = w(40 - w)$

C. $A = 20w$

D. $A = w(20 - w)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q3/5

1815

QUESTION ID

S80-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A = w(20 − w)

GRADING NOTE The length is 20 − w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q3/5

True or false?

The area of a circle as a function of its radius is $A = \pi r^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q4/5

1816

QUESTION ID

S80-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the area formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q4/5

True or false?

If a rectangle's length is its width w plus 2, then its area is w + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q5/5

1817

QUESTION ID

S80-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is w(w + 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.A.1.a

BACKWARD 2

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

In R = (20 + n)(300 − 10n), what does the factor 300 − 10n represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q1/5

1819

QUESTION ID

S80-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The number of tickets sold after n price increases of $1.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In A = x(100 − 2x) for a pen against a wall, what does 100 − 2x represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q2/5

1820

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The length of the side parallel to the wall, in meters.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In $h = -4.9t^2 + 20t + 5$, what does 5 represent?

A. The starting upward speed

B. The starting height

C. The effect of gravity

D. The time in the air

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q3/5

1821

QUESTION ID

S80-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The starting height

GRADING NOTE At t = 0, h = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In $-4.9t^2 + 20t + 5$, the coefficient of t is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q4/5

1822

QUESTION ID

S80-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The term is 20t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q4/5

1822

True or false?

The constant term of $2w^2 + 3w$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q5/5

1823

QUESTION ID

S80-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There is no constant term (it is 0); 3 is the coefficient of w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.A.1.a

FORWARD 1

Use quadratic models to make the best design choice • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

For a pen with area A(x) = x(100 − 2x), which x gives the greatest area, and what is that area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q1/5

1825

QUESTION ID

S80-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 25 m; 1,250 m²

GRADING NOTE Form required: x and the area. The zeros are 0 and 50, so the vertex is at x = 25; A(25) = 25 · 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For R(n) = (20 + n)(300 − 10n), which number of $1 increases gives the most revenue? What are the ticket price and the revenue?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q2/5

1826

QUESTION ID

S80-F1-Q2

TYPE

Short answer

CORRECT ANSWER

n = 5: a price of $25 and revenue of $6,250

GRADING NOTE Form required: n, the price, and the revenue. The zeros are −20 and 30, so the vertex is at n = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A 24-m rope is laid out as a rectangle. Which dimensions give the greatest area?

A. 2 m by 10 m

B. 12 m by 12 m

C. 6 m by 6 m

D. 4 m by 8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q3/5

1827

QUESTION ID

S80-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6 m by 6 m

GRADING NOTE Area = w(12 − w) is greatest at w = 6 (36 m²). A 12 m by 12 m rectangle would need 48 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q3/5

True or false?

Of all rectangles with perimeter 40, the square has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q4/5

1828

QUESTION ID

S80-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A = w(20 − w) is greatest at w = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q4/5

True or false?

For A(x) = x(100 − 2x), the area is greatest when x = 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S80-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q5/5

1829

QUESTION ID
S80-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A(50) = 0; the greatest area is at x = 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S80-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.A.1.a

FORWARD 2

Compose functions to build quadratic models • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

Ticket demand is $q(p) = 500 - 10p$ tickets at a price of p dollars, and the price after n increases is $p(n) = 20 + n$.

Write $q(p(n))$ and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q1/5

1831

QUESTION ID

S80-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$q(p(n)) = 300 - 10n$: the number of tickets sold after n price increases.

GRADING NOTE Form required: the simplified composite and its meaning. $500 - 10(20 + n)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square's area is $A(s) = s^2$, and its side grows with time: $s(t) = 2 + 0.5t$.

Write $A(s(t))$ in standard form.

$A(s(t)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q2/5

1832

QUESTION ID

S80-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$0.25t^2 + 2t + 4$

GRADING NOTE Form required: standard form. $(2 + 0.5t)^2 = 4 + 2t + 0.25t^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q2/5

$f(x) = x^2$ and $g(x) = x + 3$. Which is $f(g(x))$ in standard form?

A. $x^2 + 3x$

B. $x^2 + 9$

C. $x^2 + 3$

D. $x^2 + 6x + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q3/5

1833

QUESTION ID

S80-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x^2 + 6x + 9$

GRADING NOTE $(x + 3)^2 = x^2 + 6x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $f(x) = x^2$ and $g(x) = 2x$, then $g(f(3)) = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q4/5

1834

QUESTION ID

S80-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $f(3) = 9$ and $g(9) = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x^2$ and $g(x) = 2x$, then $f(g(3)) = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q5/5

1835

QUESTION ID

S80-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $g(3) = 6$ and $f(6) = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q5/5

Combine standard function types using arithmetic operations

MAIN	Algebra I main question	**HSF-BF.A.1.b**
BACKWARD 1	Add, subtract, and multiply polynomials	**HSA-APR.A.1**
BACKWARD 2	Evaluate and interpret functions	**HSF-IF.A.2**
FORWARD 1	Rewrite rational expressions such as average cost	**HSA-APR.D.6**
FORWARD 2	Graph and interpret an average-cost rational function	**HSF-IF.C.7.d**

SET 49 • HSF-BF.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Revenue is R(x) = 25x and cost is C(x) = 400 + 9x dollars for x items.

Write the profit function P(x) in simplest form and find P(50).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q1/5

1838

QUESTION ID
S49-M-Q1

TYPE
Short answer

CORRECT ANSWER
P(x) = 16x − 400; P(50) = 400

GRADING NOTE Form required: P(x) simplified and the value. P = R − C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -4.9t^2 + 30t + 2$ meters after t seconds.

Name the three simple functions added together, and what each one describes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q2/5

1839

QUESTION ID
S49-M-Q2

TYPE
Short answer

CORRECT ANSWER
The constant 2 (the starting height), the linear 30t (height gained from the upward speed), and the quadratic $-4.9t^2$ (the pull of gravity).

GRADING NOTE Any equivalent description of the three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2 + 1$ and $g(x) = 2x - 3$.

Find $(f - g)(x)$ in simplest form, and find $(f \cdot g)(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q3/5

1840

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(f - g)(x) = x^2 - 2x + 4$; $(f \cdot g)(2) = 5$

GRADING NOTE Form required: the simplified difference and the value. $f(2) = 5$ and $g(2) = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q3/5

The temperature of tea is $T(t) = 20 + 70(0.9)^t$ °C after t minutes.

Which description fits this combination of functions?

A. A linear function with slope 0.9

B. An exponential function that grows 90% per minute

C. A constant room temperature of 20 °C plus a difference of $70(0.9)^t$ that decays

D. A constant temperature of 90 °C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q4/5

1841

QUESTION ID

S49-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A constant room temperature of 20 °C plus a difference of $70(0.9)^t$ that decays

GRADING NOTE The constant 20 is added to an exponential decay term that starts at 70.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q4/5

True or false?

If $f(x) = 3x$ and $g(x) = x^2$, then $(f + g)(2) = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q5/5

1842

QUESTION ID

S49-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $f(2) + g(2) = 6 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 1

Add, subtract, and multiply polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Subtract.

$25x - (400 + 9x)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q1/5

1844

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16x − 400

GRADING NOTE Form required: simplified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$(x^2 + 1) + (2x - 3)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q2/5

1845

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x - 2$

GRADING NOTE Form required: simplified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(x^2 + 1)(2x - 3)$

A. $2x^3 + 2x - 3$

B. $2x^2 + 2x - 3$

C. $2x^3 - 3x^2 - 3$

D. $2x^3 - 3x^2 + 2x - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q3/5

1846

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2x^3 - 3x^2 + 2x - 3$

GRADING NOTE Multiply each term of the first by each term of the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

The product of two polynomials is a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q4/5

1847

QUESTION ID

S49-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Polynomials are closed under multiplication.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

$(x + 2) - (x - 2) = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q5/5

1848

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 2) - (x - 2) = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 2

Evaluate and interpret functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$P(x) = 16x - 400.$

Find x so that $P(x) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q1/5

1850

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$T(t) = 20 + 70(0.9)^t$.

Find T(0).

T(0) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q2/5

1851

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90

GRADING NOTE Any equivalent form. $(0.9)^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

R(x) is the revenue in dollars from selling x items.

What does R(10) = 250 mean?

A. Selling 250 items brings in $10.

B. Each item costs $10.

C. The profit is $250.

D. Selling 10 items brings in $250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q3/5

1852

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Selling 10 items brings in $250.

GRADING NOTE Input: items. Output: dollars of revenue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q3/5

True or false?

If C(x) = 400 + 9x, then C(0) = 400.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q4/5

1853

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9 \cdot 0 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q4/5

True or false?

If $h(t) = -4.9t^2 + 30t + 2$, then $h(0) = 30$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q5/5

1854

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $h(0) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 1

Rewrite rational expressions such as average cost • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Write the average cost A(x) = (400 + 9x) ÷ x as a constant plus a fraction.

A(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q1/5

1856

QUESTION ID
S49-F1-Q1

TYPE
Short answer

CORRECT ANSWER
9 + 400 ÷ x

GRADING NOTE Form required: a constant plus a fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q1/5

Assume $x \neq -1$. Write $(x^2 + 3x + 5) \div (x + 1)$ in the form $q(x) + r \div (x + 1)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q2/5

1857

QUESTION ID
S49-F1-Q2

TYPE
Short answer

CORRECT ANSWER
x + 2 + 3 ÷ (x + 1)

GRADING NOTE Form required: $q(x) + r \div (x + 1)$. $x^2 + 3x + 5 = (x + 1)(x + 2) + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq 0$. Which expression equals $(6x^2 + 4x) \div (2x)$?

A. $3x + 4x$

B. $3x^2 + 2$

C. $6x + 2$

D. $3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q3/5

1858

QUESTION ID
S49-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 3x + 2

GRADING NOTE Divide each term by 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q3/5

True or false?

$(x^2 - 1) \div (x - 1) = x + 1$ for $x \neq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q4/5

1859

QUESTION ID
S49-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $x^2 - 1 = (x - 1)(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x + 8) \div x = 9$ for $x \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q5/5

1860

QUESTION ID
S49-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 8) \div x = 1 + 8 \div x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 2

Graph and interpret an average-cost rational function • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

A(x) = 9 + 400 ÷ x is the average cost per item, in dollars, for x items.

Give its horizontal asymptote and what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q1/5

1862

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 9: for very many items, the average cost gets close to $9.

GRADING NOTE Form required: the asymptote and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q1/5

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items.

Find A(100).

A(100) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q2/5

1863

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 9 + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q2/5

What is the vertical asymptote of y = 400 ÷ x + 9?

A. x = 400

B. x = 9

C. y = 9

D. x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q3/5

1864

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 0

GRADING NOTE The function is undefined at x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q3/5

True or false?

As x increases, 9 + 400 ÷ x decreases toward 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q4/5

1865

QUESTION ID
S49-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

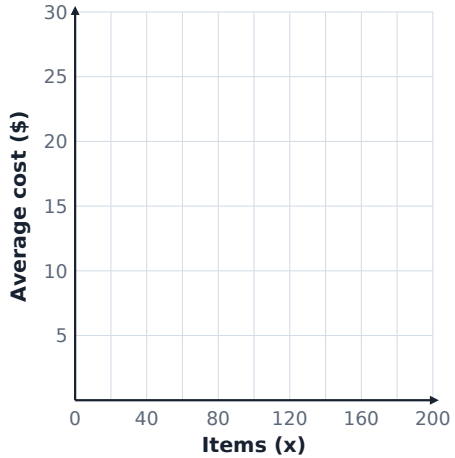
GRADING NOTE 400 ÷ x shrinks toward 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A(x) = 9 + 400 ÷ x is the average cost per item, in dollars, for x items. Graph A for 20 ≤ x ≤ 200, and draw its horizontal asymptote.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q5/5

1866

QUESTION ID

S49-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A decreasing curve through (20, 29), (40, 19), (50, 17), (100, 13), and (200, 11), approaching the line y = 9 from above

GRADING NOTE Any correct graph with the asymptote y = 9 drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q5/5

Write arithmetic and geometric sequences recursively and with an explicit formula

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate recursively defined sequences	**HSF-IF.A.3**
BACKWARD 2	Generate a pattern from a rule and describe its features	**4.OA.C.5**
FORWARD 1	Sum geometric sequences	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find a term number	**HSF-LE.A.4**

SET 50 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula and a recursive formula for 7, 4, 1, −2, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q1/5

1869

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

a(n) = 10 − 3n;
a(1) = 7 and
a(n) = a(n − 1) − 3

GRADING NOTE Form required: both formulas. The explicit formula may be written 7 − 3(n − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula and a recursive formula for 3, 12, 48, 192, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q2/5

1870

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

$a(n) = 3 \cdot 4^{n-1}$;
$a(1) = 3$ and
$a(n) = 4a(n - 1)$

GRADING NOTE Form required: both formulas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An arithmetic sequence has a(3) = 14 and a(8) = 39.

Find a(1) and the common difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q3/5

1871

QUESTION ID

S50-M-Q3

TYPE

Short answer

CORRECT ANSWER

a(1) = 4; d = 5

GRADING NOTE Form required: both. 5 steps add 25, so d = 5, and a(1) = 14 − 2(5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A geometric sequence of real numbers has a(1) = 2 and a(4) = 54.

What is the common ratio?

A. 3

B. 18

C. 27

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q4/5

1872

QUESTION ID

S50-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE $2r^3 = 54$, so $r^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The sequence 2, 4, 8, 14, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q5/5

1873

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences are 2, 4, 6, which are not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 1

Evaluate recursively defined sequences • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 4$ and $f(n) = f(n - 1) + 6$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q1/5

1875

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form. 4, 10, 16, 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 1$ and $f(n) = 2f(n - 1) + 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q2/5

1876

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 1, 3, 7, 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 2n - 1$.

What is $f(10)$?

A. 20

B. 21

C. 19

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q3/5

1877

GRADING NOTE $2(10) - 1$.

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A recursive formula defines each term after the starting term(s) using one or more earlier terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q4/5

1878

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(n) = f(n − 1) + 6 uses one earlier term; the rule f(n + 1) = f(n) + f(n − 1) uses two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q4/5

True or false?

f(1) = 3 and f(n) = 2f(n − 1) give 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q5/5

1879

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They give 3, 6, 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 2

Generate a pattern from a rule and describe its features • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Rule: start at 5 and add 5.

Write the first 6 terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q1/5

1881

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5, 10, 15, 20, 25, 30

GRADING NOTE Form required: the six terms in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rule: start at 1 and add 4.

Explain why every term is odd.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q2/5

1882

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The first term is odd, and adding 4 (an even number) to an odd number gives an odd number.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rule: start at 2 and multiply by 2.

What is the 5th term?

A. 10

B. 16

C. 64

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q3/5

1883

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 32

GRADING NOTE 2, 4, 8, 16, 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q3/5

1883

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

With the rule "start at 3 and add 3," every term is a multiple of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q4/5

1884

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3, 6, 9, 12, ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

With the rule "start at 10 and add 10," the 4th term is 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q5/5

1885

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10, 20, 30, 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum of the first 6 terms of 3, 12, 48, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q1/5

1887

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4,095

GRADING NOTE Any equivalent form. $3(4^6 - 1) \div 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A job pays $1 on day 1, and the pay doubles each day.

What is the total pay for 10 days?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q2/5

1888

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$1,023

GRADING NOTE Any equivalent form. $2^{10} - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 5 + 10 + 20 + 40 + 80?

A. 160

B. 80

C. 155

D. 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q3/5

1889

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 155

GRADING NOTE $5(2^5 - 1) = 155$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q3/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ works for $r = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q4/5

1890

QUESTION ID

S50-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It works for every $r \neq 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q4/5

True or false?

$2 + 6 + 18 + 54 = 2(3^4 - 1)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q5/5

1891

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is $80 = 2(3^4 - 1) \div 2$, not 160.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 2

Use logarithms to find a term number • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$a(n) = 3 \cdot 4^{n-1}$.

What is the smallest n for which a(n) > 10,000?

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

1893

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. $4^{n-1} > 3{,}333.3...$ gives $n - 1 > \log 3{,}333.3 \div \log 4 \approx 5.85$. Check: a(6) = 3,072 and a(7) = 12,288.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sequence starts at 5 and grows by 20% per term: $a(n) = 5(1.2)^{n-1}$. Which term is the first to exceed 100? (log means base 10.)

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

1894

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form. $1.2^{n-1} > 20$ gives $n - 1 > \log 20 \div \log 1.2 \approx 16.4$. Check: $a(17) \approx 92.4$ and $a(18) \approx 110.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

Solve $5 \cdot 2^n = 600$. (log means base 10.)

A. n = 7

B. n = log 600 ÷ log 2 ≈ 9.23

C. n = log 120 ÷ log 2 ≈ 6.91

D. n = 120 ÷ 2 = 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

1895

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n = log 120 ÷ log 2 ≈ 6.91

GRADING NOTE $2^n = 120$, so n = log 120 ÷ log 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

If $4^k = 64$, then k = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

1896

QUESTION ID

S50-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

If $3^n = 30$, then n = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

1897

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3^{10} is much larger than 30; n = log 30 ÷ log 3 ≈ 3.10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

Model situations with arithmetic and geometric sequences and translate between recursive and explicit forms

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate sequences from recursive and explicit rules	**HSF-IF.A.3**
BACKWARD 2	Write a function or a recursive process from a situation	**HSF-BF.A.1.a**
FORWARD 1	Sum geometric sequences that model situations	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find when a sequence reaches a value	**HSF-LE.A.4**

SET 81 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges $50 to join and $30 each month. a(n) is the total paid after n months, with a(1) = 80.

Write a recursive rule and an explicit rule for a(n).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q1/5

1900

QUESTION ID

S81-M-Q1

TYPE

Short answer

CORRECT ANSWER

a(1) = 80 and
a(n) = a(n − 1) + 30
for n ≥ 2;
a(n) = 50 + 30n

GRADING NOTE Form required: both rules. The explicit rule may be written 80 + 30(n − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball bounces to 80% of its previous height. Its first bounce reaches 5 m.

Write an explicit rule and a recursive rule for the height h(n) of the nth bounce.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q2/5

1901

QUESTION ID

S81-M-Q2

TYPE

Short answer

CORRECT ANSWER

$h(n) = 5(0.8)^{n-1}$; $h(1) = 5$ and $h(n) = 0.8h(n - 1)$ for $n \geq 2$

GRADING NOTE Form required: both rules.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Translate the explicit rule a(n) = 7 + 4(n − 1), for n ≥ 1, into a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q3/5

1902

QUESTION ID

S81-M-Q3

TYPE

Short answer

CORRECT ANSWER

a(1) = 7 and
a(n) = a(n − 1) + 4
for n ≥ 2

GRADING NOTE Form required: the starting value and the recursion with its condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which explicit rule matches $a(1) = 3$ and $a(n) = 2a(n - 1)$ for $n \geq 2$?

A. $a(n) = 3 \cdot 2^{n-1}$

B. $a(n) = 2 \cdot 3^{n-1}$

C. $a(n) = 3 \cdot 2^{n}$

D. $a(n) = 3 + 2(n - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q4/5

1903

QUESTION ID

S81-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $a(n) = 3 \cdot 2^{n-1}$

GRADING NOTE Each term doubles, starting from 3: 3, 6, 12, ... $3 \cdot 2^{n}$ gives 6 for $n = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q4/5

True or false?

The recursive rule a(n) = a(n − 1) − 6 needs a starting value, such as a(1) = 50, to define a sequence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q5/5

1904

QUESTION ID

S81-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Without a starting value, many sequences fit the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q5/5

SET 81 • HSF-BF.A.2

BACKWARD 1

Evaluate sequences from recursive and explicit rules • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 50$ and $f(n) = f(n - 1) - 6$ for $n \geq 2$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q1/5

1906

QUESTION ID

S81-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form. 50, 44, 38, 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 5(0.8)^{n-1}$.

Find f(3).

f(3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q2/5

1907

QUESTION ID

S81-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.2

GRADING NOTE Any equivalent form. 5(0.64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q2/5

$f(1) = 2$ and $f(n) = 3f(n - 1) + 1$ for $n \geq 2$.

What is $f(3)$?

A. 7

B. 22

C. 19

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q3/5

1908

QUESTION ID

S81-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 22

GRADING NOTE $f(2) = 7$ and $f(3) = 22$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The domain of the sequence a(n) = 2n + 1, n ≥ 1, is the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q4/5

1909

QUESTION ID

S81-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A sequence is defined at term numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

$f(n) = n^2$ gives the sequence 1, 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q5/5

1910

QUESTION ID

S81-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives 1, 4, 9, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-BF.A.2

BACKWARD 2

Write a function or a recursive process from a situation • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A savings jar starts with $20 and gains $5 each week.

Write a function for the amount A after w weeks.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q1/5

1912

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20 + 5w

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q1/5

A culture of 100 cells triples every day.

Write a function for the number of cells N after d days.

N = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q2/5

1913

QUESTION ID

S81-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$100 \cdot 3^d$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which recursive process means "start at 10 and multiply by 1.1 each year"?

A. $a(0) = 10, a(n) = a(n - 1) + 1.1$

B. $a(n) = 10 + 1.1n$

C. $a(0) = 1.1, a(n) = 10a(n - 1)$

D. $a(0) = 10, a(n) = 1.1a(n - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q3/5

1914

QUESTION ID

S81-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $a(0) = 10$, $a(n) = 1.1a(n - 1)$

GRADING NOTE Each year's amount is 1.1 times the year before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q3/5

True or false?

Adding the same amount at each step gives a linear function of the step number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q4/5

1915

QUESTION ID

S81-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equal differences mean linear growth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q4/5

True or false?

Multiplying by 2 at each step, starting at 1, gives f(n) = 2n.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q5/5

1916

QUESTION ID

S81-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives f(n) = 2^n: 1, 2, 4, 8, ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences that model situations • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A ball's bounce heights are 5, 4, 3.2, ... meters (each 80% of the one before). Find the total of the first 5 bounce heights, to the nearest hundredth.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q1/5

1918

QUESTION ID

S81-F1-Q1

TYPE

Short answer

CORRECT ANSWER

16.81 m

GRADING NOTE Form required: rounded to the nearest hundredth. $5(1 - 0.8^5) \div (1 - 0.8) = 25(0.67232) = 16.808$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

You deposit $100 the first month and 3% more each month after that. Find the total deposited in 12 months, to the nearest cent.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q2/5

1919

QUESTION ID

S81-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$1,419.20

GRADING NOTE Form required: to the nearest cent. $100(1.03^{12} - 1) \div 0.03 \approx 1{,}419.20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q2/5

Which expression is the sum of the first 10 terms of $5(0.8)^{n-1}$?

A. $5(1 - 0.8^{10}) \div 0.2$

B. $5(0.8)^{10}$

C. $10(5)(0.8)$

D. $5 \div 0.2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q3/5

1920

QUESTION ID

S81-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $5(1 - 0.8^{10}) \div 0.2$

GRADING NOTE $a(1 - r^n) \div (1 - r)$ with $a = 5$ and $r = 0.8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q3/5

True or false?

The sum of the first n bounce heights 5, 4, 3.2, ... is always less than 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q4/5

1921

QUESTION ID

S81-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The sum is $25(1 - 0.8^n)$, which is less than 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q4/5

True or false?

For a geometric series with ratio 2, adding one more term always doubles the sum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q5/5

1922

QUESTION ID

S81-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1 + 2 = 3, but 1 + 2 + 4 = 7, not 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-BF.A.2

FORWARD 2

Use logarithms to find when a sequence reaches a value • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$h(n) = 5(0.8)^{n-1}$ is the height of the nth bounce, in meters. Which bounce is the first below 1 m? (log means base 10.)

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q1/5

1924

QUESTION ID

S81-F2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $0.8^{n-1} < 0.2$ gives $n - 1 > \log 0.2 \div \log 0.8 \approx 7.2$. Check: $h(8) \approx 1.05$ and $h(9) \approx 0.84$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$a(n) = 1{,}000(1.05)^n$ is a savings balance after n years. After how many whole years does it first exceed $2,000? (log means base 10.)

Years: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q2/5

1925

QUESTION ID

S81-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. $n > \log 2 \div \log 1.05 \approx 14.2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3^n = 50$. (log means base 10.)

A. n = log 3 ÷ log 50

B. n = 50 ÷ 3

C. n = log 50 ÷ log 3 ≈ 3.56

D. n = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q3/5

1926

QUESTION ID

S81-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n = log 50 ÷ log 3 ≈ 3.56

GRADING NOTE Take the logarithm of both sides: n log 3 = log 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q3/5

True or false?

$\log(0.8^n) = n \log 0.8$, and $\log 0.8$ is negative. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q4/5

1927

QUESTION ID

S81-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.8 < 1$, so its logarithm is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $2^n = 10$, then n is a whole number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q5/5

1928

QUESTION ID

S81-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^3 = 8$ and $2^4 = 16$, so $n \approx 3.32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q5/5

Identify the effect of replacing f(x) by f(x) + k, kf(x), f(kx), and f(x + k), and recognize even and odd functions

MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph absolute value and square root functions	**HSF-IF.C.7.b**
BACKWARD 2	Read key features of quadratic graphs	**HSF-IF.C.7.a**
FORWARD 1	Describe a sequence of rigid transformations	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs	**HSF-TF.B.5**

SET 51 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

$g(x) = (x - 4)^2 + 1$.

Describe how to get the graph of g from the graph of $y = x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q1/5

1931

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

Shift it 4 units right and 1 unit up.

GRADING NOTE Any equivalent description of both shifts.

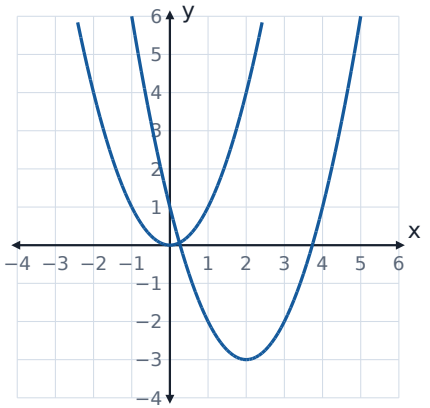
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $y = x^2$ and a translated graph $y = g(x)$.

Write g(x).



g(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q2/5

1932

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 - 3$

GRADING NOTE Any equivalent form, such as $x^2 - 4x + 1$. The vertex moved from (0, 0) to (2, −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$. Compare $g(x) = 3f(x)$ and $h(x) = f(3x)$.

Which is a vertical stretch and which is a horizontal compression? Are g and h the same function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q3/5

1933

QUESTION ID

S51-M-Q3

TYPE

Short answer

CORRECT ANSWER

$g(x) = 3x^2$ is a vertical stretch by 3; $h(x) = 9x^2$ is a horizontal compression by a factor of $\frac{1}{3}$. They are not the same.

GRADING NOTE Form required: both descriptions and the comparison. $(3x)^2 = 9x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q3/5

$f(x) = x^3$.

Is f even, odd, or neither?

A. Odd: $f(-x) = -f(x)$

B. Even: $f(-x) = f(x)$

C. Both even and odd

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q4/5

1934

QUESTION ID

S51-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Odd: $f(-x) = -f(x)$

GRADING NOTE $(-x)^3 = -x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q4/5

True or false?

The graph of y = f(x + 2) is the graph of y = f(x) shifted 2 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q5/5

1935

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted 2 units left: the output at x is f's output at x + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 1

Graph absolute value and square root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

What is the vertex of $y = |x + 3| - 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q1/5

1937

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give two points on the graph of $y = \sqrt{(x - 1)}$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q2/5

1938

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

For example (1, 0) and (5, 2)

GRADING NOTE Any two points with $y = \sqrt{(x - 1)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q2/5

What is the vertex of $y = -|x - 1| + 4$?

A. (1, 4)

B. (1, −4)

C. (4, 1)

D. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q3/5

1939

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 4)

GRADING NOTE The expression inside is 0 at x = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q3/5

True or false?

y = |x| has its minimum at (0, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q4/5

1940

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE |x| ≥ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q4/5

True or false?

The domain of $y = \sqrt{(x + 2)}$ is $x \geq 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q5/5

1941

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x + 2 \geq 0$ gives $x \geq -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 2

Read key features of quadratic graphs • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Give the vertex of $y = (x - 2)^2 - 3$, and say whether it is a maximum or a minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q1/5

1943

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(2, −3), a minimum

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B2-Q1**

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the x-intercepts of $y = x^2 - 9$.

x-intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q2/5

1944

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 0) and (−3, 0)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B2-Q2**

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the y-intercept of $y = 2x^2 - 5x + 7$?

A. 7

B. 0

C. −5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q3/5

1945

QUESTION ID

S51-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7

GRADING NOTE Substitute x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = -x^2 + 4$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q4/5

1946

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The coefficient of x^2 is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q4/5

True or false?

The vertex of $y = x^2 + 6$ is (6, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q5/5

1947

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The vertex is (0, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSF-BF.B.3

FORWARD 1

Describe a sequence of rigid transformations • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Describe a transformation that maps the triangle with vertices (0, 0), (2, 0), (0, 1) to the triangle with vertices (3, 2), (5, 2), (3, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q1/5

1949

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A translation by (x, y) → (x + 3, y + 2).

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q1/5

The point (2, 5) is reflected across the x-axis and then translated 4 units left.

Where does it end up?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q2/5

1950

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

GRADING NOTE Any equivalent form. (2, 5) → (2, −5) → (−2, −5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which single transformation maps every (x, y) to (−x, −y)?

A. A translation

B. A rotation of 180° about the origin

C. A reflection across the y-axis

D. A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q3/5

1951

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A rotation of 180° about the origin

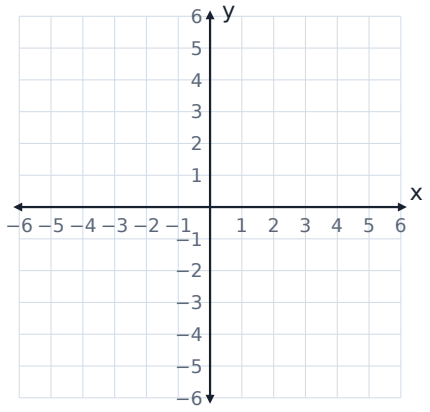
GRADING NOTE Both coordinates change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(1, 1), B(4, 1), and C(1, 3). Reflect it across the y-axis, then translate the image 2 units down. Draw the final image and label its vertices.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q4/5

1952

QUESTION ID

S51-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with A'(−1, −1), B'(−4, −1), and C'(−1, 1)

GRADING NOTE Any correct drawing. Reflecting across the y-axis gives (−1, 1), (−4, 1), (−1, 3); then subtract 2 from each y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q4/5

True or false?

Translating and then reflecting always gives the same result as reflecting and then translating.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q5/5

1953

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For (1, 0), translating right 1 then reflecting across the y-axis gives (−2, 0); reflecting first then translating gives (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of y = sin x + 3 related to the graph of y = sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q1/5

1955

QUESTION ID
S51-F2-Q1

TYPE
Short answer

CORRECT ANSWER
It is shifted 3 units up; its midline is y = 3.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a sine function with amplitude 2, midline y = 1, and period 2π whose graph passes through (0, 1) and is rising there.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q2/5

1956

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2sin x + 1

GRADING NOTE Any equivalent form, such as 2sin(x + 2π) + 1. The conditions at (0, 1) fix the phase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How is y = 4sin x obtained from y = sin x?

A. A shift 4 units right

B. A horizontal stretch by a factor of 4

C. A vertical stretch by a factor of 4

D. A shift 4 units up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q3/5

1957

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A vertical stretch by a factor of 4

GRADING NOTE Every output is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = -\sin x$ is the reflection of $y = \sin x$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q4/5

1958

QUESTION ID
S51-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Every output changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = \sin x - 2$ has midline $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q5/5

1959

QUESTION ID
S51-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its midline is $y = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q5/5

Experiment with transformations of graphs and recognize even and odd functions from graphs and equations

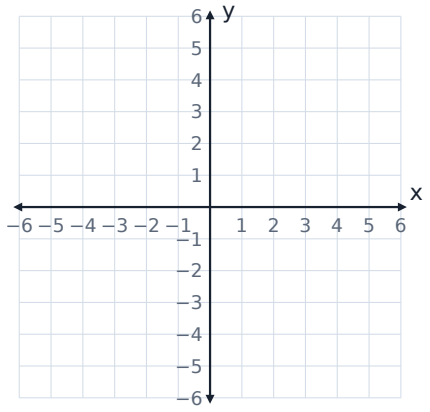
MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph quadratic functions	**HSF-IF.C.7.a**
BACKWARD 2	Graph absolute value and root functions	**HSF-IF.C.7.b**
FORWARD 1	Draw rigid transformations of figures	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs and recognize their symmetry	**HSF-TF.B.5**

SET 82 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

Graph $f(x) = x^2$ and $g(x) = (x - 3)^2 - 2$ on the same grid. Label both, and describe how g is related to f.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q1/5

1962

QUESTION ID

S82-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Both parabolas; g has vertex (3, −2): it is f shifted 3 units right and 2 units down

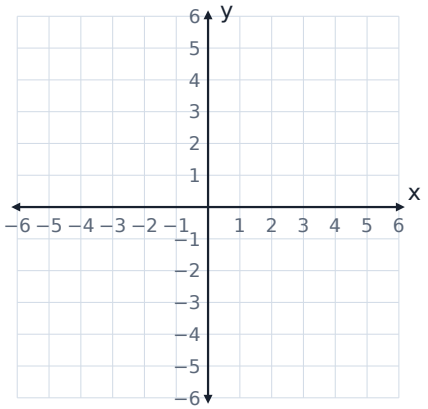
GRADING NOTE Any correct graphs with the shift described.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = |x|$. Graph $g(x) = 2f(x)$ and $h(x) = f(2x)$ for $-3 \le x \le 3$, using a table or graphing technology. Are g and h the same function?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q2/5

1963

QUESTION ID

S82-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: One V-shape with vertex (0, 0) and slopes 2 and −2 (both g and h)

Written: Yes. $|2x| = 2|x|$ for every x, so g and h have the same graph.

GRADING NOTE Form required: the answer with a reason or table. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a table or graphing technology to compare $f(x) = \sqrt{x}$ with $g(x) = \sqrt{(x - 4)} + 1$ at $x = 4$, 5, and 8. Describe the transformation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q3/5

1964

QUESTION ID

S82-M-Q3

TYPE

Short answer

CORRECT ANSWER

g(4) = 1, g(5) = 2, g(8) = 3, which match f(0) = 0, f(1) = 1, f(4) = 2 moved 4 right and 1 up: g is f shifted 4 units right and 1 unit up.

GRADING NOTE Form required: the values and the transformation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of f is shown.

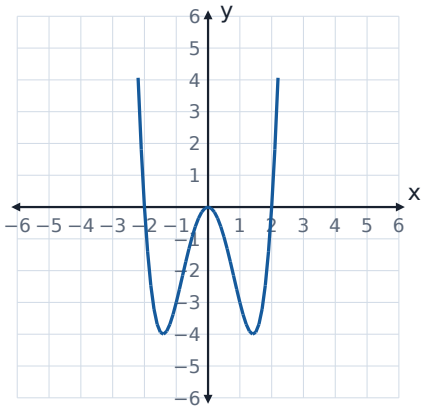
Is f even, odd, or neither?

A. Both even and odd

B. Even

C. Odd

D. Neither



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q4/5

1965

QUESTION ID

S82-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Even

GRADING NOTE The graph is symmetric about the y-axis, so f(−x) = f(x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q4/5

True or false?

$f(x) = x^3 + x$ is an odd function, because $f(-x) = -f(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q5/5

1966

QUESTION ID

S82-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-x)^3 + (-x) = -(x^3 + x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q5/5

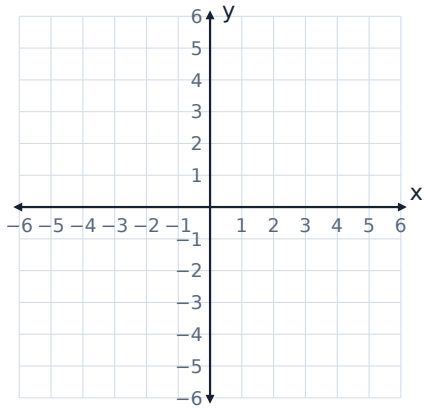
SET 82 • HSF-BF.B.3

BACKWARD 1

Graph quadratic functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y = x^2 - 4$, and mark its vertex and x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q1/5

1968

QUESTION ID

S82-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A parabola opening up with vertex (0, −4), crossing the x-axis at (−2, 0) and (2, 0)

GRADING NOTE Any correct graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the vertex of $y = (x + 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q2/5

1969

QUESTION ID

S82-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q2/5

What is the axis of symmetry of $y = x^2 - 6x + 1$?

A. $x = 6$

B. $x = 3$

C. $x = -3$

D. $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q3/5

1970

QUESTION ID

S82-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 3

GRADING NOTE $x = -b \div (2a) = 6 \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = -x^2 + 1$ has a maximum value of 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q4/5

1971

QUESTION ID

S82-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-x^2 \leq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q4/5

True or false?

The x-intercepts of $y = x^2 + 1$ are 1 and −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S82-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q5/5

1972

QUESTION ID

S82-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 + 1$ is never 0, so there are no x-intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S82-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-BF.B.3

BACKWARD 2

Graph absolute value and root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

What is the vertex of $y = |x - 4| + 1$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q1/5

1974

QUESTION ID

S82-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 1)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Where does the graph of $y = \sqrt{(x + 2)} - 3$ start?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q2/5

1975

QUESTION ID

S82-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function's graph is V-shaped?

A. $y = \sqrt[3]{(x + 1)}$

B. $y = |x + 1|$

C. $y = \sqrt{(x + 1)}$

D. $y = (x + 1)^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q3/5

1976

QUESTION ID

S82-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = |x + 1|

GRADING NOTE Absolute value graphs are V-shaped.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = \sqrt[3]{x}$ passes through $(-1, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q4/5

1977

QUESTION ID

S82-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-1)^3 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q4/5

True or false?

The graph of $y = |x| - 2$ has vertex (2, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q5/5

1978

QUESTION ID

S82-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its vertex is (0, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

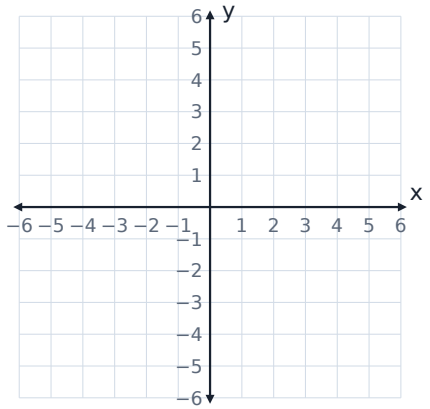
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSF-BF.B.3

FORWARD 1

Draw rigid transformations of figures • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Translate triangle A(0, 0), B(2, 0), C(0, 3) by (x, y) → (x − 3, y + 1). Draw and label the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q1/5

1980

QUESTION ID

S82-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle A'(−3, 1), B'(−1, 1), C'(−3, 4)

GRADING NOTE Any correct drawing with labels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q1/5

The point (4, −1) is rotated 90° counterclockwise about the origin. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q2/5

1981

QUESTION ID
S82-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(1, 4)

GRADING NOTE Any equivalent form. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sequence of transformations maps (x, y) to (−x, y − 2)?

A. Reflect across the x-axis, then translate 2 units down

B. Rotate 180°, then translate 2 units up

C. Reflect across the y-axis, then translate 2 units down

D. Translate 2 units left, then reflect across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q3/5

1982

QUESTION ID
S82-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Reflect across the y-axis, then translate 2 units down

GRADING NOTE −x comes from the reflection, and y − 2 from the translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q3/5

True or false?

A translation followed by another translation is a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q4/5

1983

QUESTION ID

S82-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The two shifts add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q4/5

True or false?

Reflecting a figure across the same line twice moves it to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q5/5

1984

QUESTION ID

S82-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second reflection undoes the first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q5/5

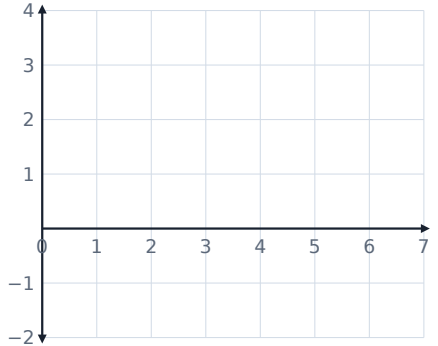
SET 82 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs and recognize their symmetry • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Sketch $y = 2\sin x + 1$ for $0 \le x \le 2\pi$ (about 6.28).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q1/5

1986

QUESTION ID

S82-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A wave starting at (0, 1), rising to 3 near x = 1.57, back to 1 near 3.14, down to −1 near 4.71, and back to 1 near 6.28

GRADING NOTE Any correct sketch with midline 1, amplitude 2, and period 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe how the graph of $y = \sin(x - \pi \div 2)$ is related to the graph of $y = \sin x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q2/5

1987

QUESTION ID

S82-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It is the graph of $y = \sin x$ shifted $\pi \div 2$ units to the right.

GRADING NOTE Any equivalent description (it is also the graph of $y = -\cos x$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function has period π?

A. y = sin x + 2

B. y = sin(x + 2)

C. y = 2sin x

D. y = sin(2x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q3/5

1988

QUESTION ID

S82-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = sin(2x)

GRADING NOTE 2π ÷ 2 = π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q3/5

True or false?

y = −cos x is the reflection of y = cos x across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q4/5

1989

QUESTION ID
S82-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Every output changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = cos x is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q5/5

1990

QUESTION ID

S82-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE cos(−x) = cos x, so cos is even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q5/5

Solve f(x) = c for a simple function f and write an expression for the inverse

MAIN	Algebra I main question	**HSF-BF.B.4.a**
BACKWARD 1	Rearrange a formula for another variable	**HSA-CED.A.4**
BACKWARD 2	Use cube roots to solve $x^3 = p$	**8.EE.A.2**
FORWARD 1	Verify inverses by composition	**HSF-BF.B.4.b**
FORWARD 2	Restrict a domain to produce an inverse	**HSF-BF.B.4.d**

SET 52 • HSF-BF.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3x^3 - 1$. Solve $f(x) = 80$, and write an expression for the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q1/5

1993

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 3$; $f^{-1}(x) = \sqrt[3]{((x + 1) \div 3)}$

GRADING NOTE Form required: the solution and the inverse (any equivalent expression). $3x^3 = 81$, so $x^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (2x + 3) \div (x - 4)$, for $x \neq 4$. Write an expression for the inverse.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q2/5

1994

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(4x + 3) \div (x - 2)$, for $x \neq 2$

GRADING NOTE Any equivalent form. Solve $y = (2x + 3) \div (x - 4)$ for x: $xy - 4y = 2x + 3$, so $x(y - 2) = 4y + 3$. Check: $f(0) = -0.75$ and $f^{-1}(-0.75) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

F(C) = 1.8C + 32 gives degrees Fahrenheit.

Write the inverse C(F), and find C when F = 77.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q3/5

1995

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

C(F) = (F − 32) ÷ 1.8; 25 °C

GRADING NOTE Form required: the inverse (any equivalent form) and the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 4x - 7$.

What is $f^{-1}(x)$?

A. $(x - 7) \div 4$

B. $(x + 7) \div 4$

C. $x \div 4 + 7$

D. $4x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q4/5

1996

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (x + 7) ÷ 4

GRADING NOTE Solve $y = 4x - 7$ for x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q4/5

True or false?

If $f(x) = x \div 3 + 2$, then $f^{-1}(x) = 3x - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q5/5

1997

QUESTION ID

S52-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = y \div 3 + 2$ gives $y = 3(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q5/5

SET 52 • HSF-BF.B.4.a

BACKWARD 1

Rearrange a formula for another variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = 4x - 7$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q1/5

1999

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(y + 7) \div 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve P = 2l + 2w for l.

l = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q2/5

2000

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(P − 2w) ÷ 2

GRADING NOTE Any equivalent form, such as P ÷ 2 − w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q2/5

2000

ANSWER KEY Reduced copy of the question page, with its answer

Solve d = rt for t. (r ≠ 0)

A. t = r ÷ d

B. t = d − r

C. t = d ÷ r

D. t = dr

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q3/5

2001

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. t = d ÷ r

GRADING NOTE Divide both sides by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q3/5

2001

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $y = x^3$ for x gives $x = \sqrt[3]{y}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q4/5

2002

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Take the cube root of both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q4/5

2002

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving y = 5x for x gives x = 5y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q5/5

2003

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = y ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q5/5

2003

SET 52 • HSF-BF.B.4.a

BACKWARD 2

Use cube roots to solve $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^3 = 27$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q1/5

2005

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\sqrt[3]{125}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q2/5

2006

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5^3 = 125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q2/5

2006

ANSWER KEY Reduced copy of the question page, with its answer

What is $\sqrt[3]{64}$?

A. 21.3

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q3/5

2007

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q3/5

2007

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt[3]{(-27)} = -3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q4/5

2008

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(-3)^3 = -27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q4/5

2008

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{16} = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q5/5

2009

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^2 = 16$, and $\sqrt{}$ means the nonnegative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-BF.B.4.a

FORWARD 1

Verify inverses by composition • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Show that $f(x) = 2x^3$ and $g(x) = \sqrt[3]{(x \div 2)}$ are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q1/5

2011

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$f(g(x)) = 2 \cdot (x \div 2) = x$ and $g(f(x)) = \sqrt[3]{(x^3)} = x$.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = (x + 1) ÷ (x − 1).

Find f(f(3)).

f(f(3)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q2/5

2012

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. f(3) = 2 and f(2) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q2/5

Which function is its own inverse?

A. $f(x) = 1 \div x$

B. $f(x) = 2x$

C. $f(x) = x + 1$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q3/5

2013

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $f(x) = 1 \div x$

GRADING NOTE $1 \div (1 \div x) = x$ for $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q3/5

True or false?

If f(g(x)) = x on the domain of g and g(f(x)) = x on the domain of f, then g is the inverse of f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q4/5

2014

QUESTION ID

S52-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the composition test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

f(x) = x + 3 and g(x) = x − 3 are not inverses, because f(g(1)) = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q5/5

2015

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(g(1)) = f(−2) = 1. They are inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q5/5

2015

SET 52 • HSF-BF.B.4.a

FORWARD 2

Restrict a domain to produce an inverse • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2 + 2$. Restrict f to the largest interval that contains $x = 1$ on which f is one-to-one, and give the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q1/5

2017

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$; $f^{-1}(x) = \sqrt{(x - 2)}$, for $x \geq 2$

GRADING NOTE Form required: the interval and the inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why f(x) = |x| has no inverse on all real numbers.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q2/5

2018

QUESTION ID
S52-F2-Q2

TYPE
Short answer

CORRECT ANSWER
f(3) = f(−3) = 3, so the input of 3 for an inverse would need two outputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q2/5

$f(x) = (x + 4)^2$.

What is the largest interval containing $x = 0$ on which f is one-to-one?

A. All real numbers

B. $-8 \le x \le 0$

C. $x \ge 0$

D. $x \ge -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q3/5

2019

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x \ge -4$

GRADING NOTE f is increasing for $x \ge -4$. On $-8 \le x \le 0$, $f(-8) = f(0) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ restricted to $x \le 0$ has the inverse $f^{-1}(x) = -\sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q4/5

2020

QUESTION ID
S52-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE For $x \le 0$, $-\sqrt{(x^2)} = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f(x) = x^2 + 1$ is one-to-one on all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q5/5

2021

QUESTION ID

S52-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(1) = f(-1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q5/5

Prove that linear functions grow by equal differences and exponential functions grow by equal factors over equal intervals

MAIN	Algebra I main question	**HSF-LE.A.1.a**
BACKWARD 1	Tell linear from nonlinear functions	**8.F.A.3**
BACKWARD 2	Compare average rates of change over intervals	**HSF-IF.B.6**
FORWARD 1	Derive the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to turn equal factors into equal differences	**HSF-BF.B.5**

SET 53 • HSF-LE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 3x + 5. Show that f grows by equal differences over equal intervals: find f(x + h) − f(x) in simplest form, and explain why it does not depend on x.

f(x + h) − f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q1/5

2024

QUESTION ID

S53-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

3h

GRADING NOTE REQUIRED WORK: Algebra showing f(x + h) − f(x) = 3(x + h) + 5 − (3x + 5) = 3h, and the statement that 3h depends only on the interval length h.

Form required: the simplified difference and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q1/5

2024

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 4 \cdot 2^x$. Show that g grows by equal factors over equal intervals: find $g(x + h) \div g(x)$ in simplest form, and explain why it does not depend on x.

$g(x + h) \div g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q2/5

2025

QUESTION ID

S53-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

2^h

GRADING NOTE REQUIRED WORK: Algebra showing $4 \cdot 2^{x+h} \div (4 \cdot 2^x) = 2^h$, and the statement that 2^h depends only on h.

Form required: the simplified ratio and the explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 2, 4, 6 and y = 5, 11, 17, 23.

Is the relationship linear or exponential? Explain using equal intervals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q3/5

2026

QUESTION ID
S53-M-Q3

TYPE
Short answer

CORRECT ANSWER
Linear: over each interval of 2, y increases by the same amount, 6.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 80, 40, 20, 10.

Which description is correct?

A. Linear: y decreases by 40 each time

B. Exponential: y is multiplied by $\frac{1}{2}$ each time x increases by 1

C. Linear: y decreases by 20 each time

D. Neither linear nor exponential

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q4/5

2027

QUESTION ID

S53-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Exponential: y is multiplied by 1/2 each time x increases by 1

GRADING NOTE The differences are −40, −20, −10 (not equal), but the ratios are all $\frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q4/5

2027

True or false?

A table with y-values 2, 5, 10, 17 for x = 0, 1, 2, 3 is linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q5/5

2028

QUESTION ID

S53-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences 3, 5, 7 are not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 1

Tell linear from nonlinear functions • Grade 8 • Functions • 8.F.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = x^2 + 1$ linear? Use the points (0, 1), (1, 2), and (2, 5) to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q1/5

2030

QUESTION ID
S53-B1-Q1

TYPE
Short answer

CORRECT ANSWER
No. y increases by 1 and then by 3, so the rate is not constant.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is y = 4 − 2x linear? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q2/5

2031

GRADING NOTE Form required: the answer and the reason.

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. It has the form y = mx + b with slope −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function is linear?

A. $y = 2^x$

B. $y = x^2$

C. $y = 0.5x - 3$

D. $y = 1 \div x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q3/5

2032

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 0.5x - 3$

GRADING NOTE Only $y = 0.5x - 3$ has the form $y = mx + b$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q3/5

True or false?

The graph of a linear function is a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q4/5

2033

QUESTION ID

S53-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is why it is called linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The area of a square, $A = s^2$, is a linear function of s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q5/5

2034

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its graph curves; the area grows by 1, 3, 5, ... for s = 0, 1, 2, 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 2

Compare average rates of change over intervals • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3x + 5$.

Find the average rate of change from $x = 0$ to $x = 10$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q1/5

2036

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 4 \cdot 2^x$.

Find the average rate of change from $x = 0$ to $x = 1$, and from $x = 1$ to $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q2/5

2037

QUESTION ID
S53-B2-Q2

TYPE
Short answer

CORRECT ANSWER
4 and 8

GRADING NOTE Form required: both rates. (8 − 4) ÷ 1 and (16 − 8) ÷ 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the average rate of change of $5 \cdot 3^x$ from $x = 0$ to $x = 2$?

A. 15

B. 20

C. 40

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q3/5

2038

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20

GRADING NOTE $(45 - 5) \div 2 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An exponential growth function has a larger average rate of change on later intervals of the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q4/5

2039

QUESTION ID

S53-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its outputs, and their changes, keep multiplying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q4/5

True or false?

The average rate of change of 2^x from $x = 0$ to $x = 3$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q5/5

2040

QUESTION ID
S53-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(8 - 1) \div 3 = \frac{7}{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSF-LE.A.1.a

FORWARD 1

Derive the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Let $S = 1 + 2 + 4 + \ldots + 2^9$. Show that $2S - S = 2^{10} - 1$, and find S.

S = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q1/5

2042

QUESTION ID

S53-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

1,023

GRADING NOTE REQUIRED WORK: 2S written as $2 + 4 + \ldots + 2^{10}$, then subtracting S so that every term except 2^{10} and 1 cancels.

Form required: the subtraction and the value of S.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q1/5

Use $S = a(1 - r^n) \div (1 - r)$ to find the sum of the first 6 terms of 1, 3, 9, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q2/5

2043

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

364

GRADING NOTE Any equivalent form. $(1 - 729) \div (1 - 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q2/5

When deriving the sum formula, why do we multiply S by r?

A. To double the sum

B. So that S − rS cancels all but two terms

C. To make every term equal

D. To make r equal 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q3/5

2044

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. So that S − rS cancels all but two terms

GRADING NOTE S and rS share every term except the first term of S and the last term of rS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For $r \neq 1$, $a + ar + \ldots + ar^{n-1} = a(1 - r^n) \div (1 - r)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q4/5

2045

QUESTION ID

S53-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the geometric series formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q4/5

True or false?

$1 + 3 + 9 + 27 = 3^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q5/5

2046

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is 40, and $3^4 = 81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSF-LE.A.1.a

FORWARD 2

Use logarithms to turn equal factors into equal differences • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 4 \cdot 2^x$.

Show that $\log_2 g(x) = x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q1/5

2048

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\log_2(4 \cdot 2^x) = \log_2$

$4 + \log_2 2^x = 2 + x.$

GRADING NOTE Form required: the steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 10^x$.

What is log y? (log means base 10.)

log y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q2/5

2049

QUESTION ID
S53-F2-Q2

TYPE
Short answer

CORRECT ANSWER

x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The values y = 1, 10, 100, 1,000 occur at x = 0, 1, 2, 3.

What are the values of log y? (log means base 10.)

A. 0, 1, 2, 3

B. 0, 10, 20, 30

C. 1, 2, 3, 4

D. 1, 10, 100, 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q3/5

2050

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0, 1, 2, 3

GRADING NOTE $\log 10^k = k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For $a > 0$ and $b > 0$, taking the logarithm of $a \cdot b^x$ gives a linear function of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q4/5

2051

QUESTION ID
S53-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $\log(a \cdot b^x) = \log a + x \log b$, a linear function of x. The logarithm needs a positive input, so $a > 0$ and $b > 0$ are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log(100 \cdot 10^x) = 100x$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q5/5

2052

QUESTION ID

S53-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(100 \cdot 10^x) = 2 + x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q5/5

Recognize situations in which a quantity changes at a constant rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.b**
BACKWARD 1	Find the constant of proportionality	**7.RP.A.2.b**
BACKWARD 2	Write a linear function from a rate and an initial value	**8.F.B.4**
FORWARD 1	Relate arc length and sector area to the central angle — nearest r…	**HSG-C.B.5**
FORWARD 2	Use radian measure as arc length on a circle — nearest related	**HSF-TF.A.1**

SET 54 • HSF-LE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

A pool fills at a constant rate. At 2 p.m. it holds 1,500 gallons; at 5 p.m. it holds 3,300 gallons.

What is the rate, and how much water was in it at noon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q1/5

2055

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

600 gallons per hour; 300 gallons at noon

GRADING NOTE Form required: the rate with units and the amount. 1,800 ÷ 3 = 600, and 1,500 − 2(600) = 300.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-M-Q1**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plant is 6 cm tall on day 0 and grows 0.8 cm each day.

How tall is it on day 15?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q2/5

2056

QUESTION ID
S54-M-Q2

TYPE
Short answer

CORRECT ANSWER
18 cm

GRADING NOTE Any equivalent form. 6 + 0.8(15).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q2/5

A taxi charges the same amount for every mile.

Explain why the fare for the miles driven is a linear function of the miles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q3/5

2057

QUESTION ID

S54-M-Q3

TYPE

Short answer

CORRECT ANSWER

Each mile adds the same amount, so the fare grows by equal differences over equal intervals, which is linear growth.

GRADING NOTE Any equivalent explanation that uses the constant rate per mile.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which situation is modeled by a linear function?

A. A cell population doubles each hour.

B. A tank loses 4 liters each minute.

C. A loan grows 4% each month.

D. A car loses 15% of its value each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q4/5

2058

QUESTION ID

S54-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A tank loses 4 liters each minute.

GRADING NOTE Only the tank changes by the same amount each unit of time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q4/5

True or false?

A savings account that earns the same dollar amount of interest each year grows linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q5/5

2059

QUESTION ID
S54-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A constant amount per year is a constant rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 1

Find the constant of proportionality • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

y = 2.5x gives the cost of x pounds of apples.

What is the constant of proportionality?

Constant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q1/5

2061

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A proportional table has x = 2, 4, 6 and y = 7, 14, 21.

What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q2/5

2062

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE Any equivalent form. 7 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A cyclist rides 12 miles in 3 hours at a constant rate.

What is the constant of proportionality in miles per hour?

A. 15

B. 36

C. $\frac{1}{4}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q3/5

2063

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

GRADING NOTE 12 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = 9x, y increases by 9 whenever x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q4/5

2064

QUESTION ID

S54-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 9 is the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A proportional relationship through (5, 20) has constant of proportionality 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q5/5

2065

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $20 \div 5 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 2

Write a linear function from a rate and an initial value • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (2, 1,500) and (5, 3,300).

Find its slope and its y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q1/5

2067

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Slope 600;
y-intercept 300

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A phone battery is at 80% and drops 5 percentage points per hour.

Write a function B(t) for the charge after t hours.

B(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q2/5

2068

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

80 − 5t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function has rate of change 0.8 and initial value 6?

A. $y = 6x + 0.8$

B. $y = 0.8x + 6$

C. $y = 6.8x$

D. $y = 6(0.8)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q3/5

2069

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 0.8x + 6$

GRADING NOTE $y = mx + b$ with $m = 0.8$ and $b = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q3/5

True or false?

A function with initial value 10 and rate of change −2 is y = 10 − 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q4/5

2070

QUESTION ID

S54-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE b = 10 and m = −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q4/5

True or false?

The line through (0, 4) and (2, 10) has rate of change 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q5/5

2071

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(10 - 4) \div 2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-LE.A.1.b

FORWARD 1

Relate arc length and sector area to the central angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: Arc length grows by a constant amount (the radius) for each radian, a constant-rate situation.

ANSWER KEY Reduced copy of the question page, with its answer

In a circle of radius 5 cm, by how much does the arc length increase for each 1-radian increase in the central angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

2073

QUESTION ID
S54-F1-Q1

TYPE
Short answer

CORRECT ANSWER
5 cm

GRADING NOTE Any equivalent form. $s = 5\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 10.

Find the arc lengths for central angles of 0.5, 1, and 1.5 radians.

Arc lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

2074

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5, 10, and 15

GRADING NOTE Form required: all three, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

The area of a sector is $A = \frac{1}{2}r^2\theta$, and $r = 4$.

By how much does A increase when θ increases by 1 radian?

A. 4

B. 2

C. 16

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

2075

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

GRADING NOTE $\frac{1}{2}(16)(1) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For a fixed radius, arc length is a linear function of the central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

2076

QUESTION ID

S54-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE s = rθ has the constant rate r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Doubling the radius while keeping the angle the same doubles the area of a sector.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

2077

QUESTION ID
S54-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The area depends on r^2, so it is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 54 • HSF-LE.A.1.b

FORWARD 2

Use radian measure as arc length on a circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Algebra II standard. Connection: Radian measure is arc length per unit of radius, continuing the constant-rate arc length of FORWARD 1.

A wheel of radius 0.3 m turns through 4π radians.

How far does a point on its rim travel? Give the exact distance and the distance to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

2079

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.2π m, about 3.77 m

GRADING NOTE Form required: exact and rounded. Arc length = rθ = 0.3 × 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A central angle cuts off an arc of length 6 on a circle of radius 2.

What is its radian measure?

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

2080

QUESTION ID
S54-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE Any equivalent form. 6 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

A wheel turns at 3 radians per second.

Through about how many degrees does it turn in 2 seconds?

A. 6°

B. 172°

C. 1,080°

D. 344°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

2081

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 344°

GRADING NOTE 6 radians × (180 ÷ π) ≈ 343.8°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

One radian is the angle that cuts off an arc as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

2082

QUESTION ID

S54-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A rotation of $-\pi \div 2$ radians is a quarter turn counterclockwise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

2083

QUESTION ID

S54-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Negative angles turn clockwise; $-\pi \div 2$ is a quarter turn clockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q5**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

Recognize situations in which a quantity grows or decays by a constant percent rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.c**
BACKWARD 1	Read growth and decay rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Solve percent problems	**7.RP.A.3**
FORWARD 1	Find times in percent-rate models with logarithms	**HSF-LE.A.4**
FORWARD 2	Use natural logarithms with continuous growth	**HSF-BF.B.5**

SET 55 • HSF-LE.A.1.c

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

A car worth $24,000 loses 15% of its value each year.

Write a function V(t), and find the value after 3 years to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q1/5

2086

QUESTION ID

S55-M-Q1

TYPE

Short answer

CORRECT ANSWER

$V(t) = 24{,}000(0.85)^t$; $14,739

GRADING NOTE Form required: the function and the rounded value. $0.85^3 = 0.614125$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why a quantity that grows by 6% each year grows by equal factors over equal intervals.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q2/5

2087

QUESTION ID
S55-M-Q2

TYPE
Short answer

CORRECT ANSWER
Each year the quantity is multiplied by the same factor, 1.06, so every 1-year interval multiplies it by 1.06.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A town's population increases by 2.5% per year.

By what factor does it grow in 2 years? What is the percent increase over 2 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q3/5

2088

QUESTION ID
S55-M-Q3

TYPE
Short answer

CORRECT ANSWER
1.050625; about 5.06%

GRADING NOTE Form required: the factor and the percent. $1.025^2 = 1.050625$, slightly more than 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which situation is exponential?

A. A worker earns $15 each hour.

B. A medicine dose decreases by 20% each hour.

C. A tank gains 3 liters each second.

D. A runner covers 200 meters each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q4/5

2089

QUESTION ID

S55-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A medicine dose decreases by 20% each hour.

GRADING NOTE Only the dose changes by a constant percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q4/5

True or false?

A 10% increase followed by a 10% decrease returns a quantity to its original value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q5/5

2090

QUESTION ID
S55-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $1.1 \times 0.9 = 0.99$, a 1% loss.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 1

Read growth and decay rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

$V = 24{,}000(0.85)^t$.

What is the yearly percent change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q1/5

2092

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A 15% decrease per year

GRADING NOTE Any equivalent form. $0.85 = 1 - 0.15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$P = 500(1.025)^t$.

Is this growth or decay, and at what rate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q2/5

2093

QUESTION ID

S55-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Growth, at 2.5% per unit of t

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function shows 4% decay per year?

A. $y = 300(1.04)^t$

B. $y = 300(0.04)^t$

C. $y = 300 - 0.04t$

D. $y = 300(0.96)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q3/5

2094

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 300(0.96)^t$

GRADING NOTE $1 - 0.04 = 0.96$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 50(1.5)^t$ grows by 50% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q4/5

2095

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1.5 = 1 + 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 80(0.75)^t$ decreases by 75% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q5/5

2096

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It keeps 75%, so it decreases by 25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 2

Solve percent problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A $24,000 car loses 15% of its value.

What is its new value?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q1/5

2098

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$20,400

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 8% sales tax on $45?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q2/5

2099

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$3.60

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the price of a $60 item after a 25% discount?

A. $15

B. $45

C. $75

D. $35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q3/5

2100

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $45

GRADING NOTE 60 × 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

20% of 150 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q4/5

2101

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0.2 × 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q4/5

True or false?

A 50% increase of 40 is 60, and a 50% decrease of 60 is 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q5/5

2102

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 50% decrease of 60 is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-LE.A.1.c

FORWARD 1

Find times in percent-rate models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$V(t) = 24{,}000(0.85)^t$.

When is V = 12,000? Round to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q1/5

2104

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4.3

GRADING NOTE Form required: rounded to the nearest tenth. t = log 0.5 ÷ log 0.85 ≈ 4.27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $1.025^t = 2$. Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q2/5

2105

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

28.1

GRADING NOTE Form required: rounded to the nearest tenth. log 2 ÷ log 1.025 ≈ 28.07.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q2/5

Which expression gives the half-life of $y = (0.9)^t$? (log means base 10.)

A. 0.9 ÷ 2

B. 0.5 ÷ 0.9

C. log 0.9 ÷ log 0.5

D. log 0.5 ÷ log 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q3/5

2106

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. log 0.5 ÷ log 0.9

GRADING NOTE Solve $0.9^t = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log(0.85^t) = t \log 0.85$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q4/5

2107

QUESTION ID

S55-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The power property of logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $2^t = 5$, then $t = 2.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q5/5

2108

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^{2.5} \approx 5.66$. $t = \log 5 \div \log 2 \approx 2.32$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSF-LE.A.1.c

FORWARD 2

Use natural logarithms with continuous growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $y = e^x$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q1/5

2110

QUESTION ID

S55-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$y = \ln x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^{0.05t} = 3$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q2/5

2111

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

20 ln 3

GRADING NOTE Form required: exact. 0.05t = ln 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\ln(e^4)$? (ln means base e.)

A. 4e

B. 4

C. 1

D. e

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q3/5

2112

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE ln and e^x are inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q3/5

True or false?

e raised to the power ln 7 equals 7. (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q4/5

2113

QUESTION ID

S55-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE e^x and ln x undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q4/5

True or false?

ln 1 = 1 (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q5/5

2114

QUESTION ID
S55-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $e^0 = 1$, so ln 1 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q5/5

Construct linear and exponential functions from a graph, a description, two points, or a table

MAIN	Algebra I main question	**HSF-LE.A.2**
BACKWARD 1	Write a linear function from two points or a table	**8.F.B.4**
BACKWARD 2	Write formulas for arithmetic and geometric sequences	**HSF-BF.A.2**
FORWARD 1	Build a sinusoidal model from a description	**HSF-TF.B.5**
FORWARD 2	Find a quadratic through three points with a matrix equation	**HSA-REI.C.8**

SET 56 • HSF-LE.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function whose graph passes through (2, 11) and (6, 23).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q1/5

2117

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 5

GRADING NOTE Any equivalent form. The slope is 12 ÷ 4 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the exponential function $y = ab^x$ $(b > 0)$ whose graph passes through (1, 6) and (3, 54).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q2/5

2118

QUESTION ID
S56-M-Q2

TYPE
Short answer

CORRECT ANSWER
$2(3)^x$

GRADING NOTE Any equivalent form. $ab^3 \div (ab) = b^2 = 9$, so $b = 3$ and $a = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

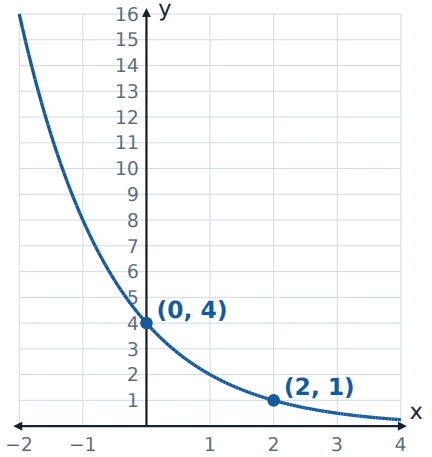
High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of an exponential function $y = ab^x$ passes through the marked points.

Write the function.

y = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q3/5

2119

QUESTION ID
S56-M-Q3

TYPE
Short answer

CORRECT ANSWER
$4(0.5)^x$

GRADING NOTE Any equivalent form, such as $4(1/2)^x$. a = 4, and $4b^2 = 1$ gives b = 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 1.5, 3, 6, 12.

Which function fits?

A. $y = 2(1.5)^x$

B. $y = 1.5x + 1.5$

C. $y = 3x$

D. $y = 1.5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q4/5

2120

QUESTION ID

S56-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 1.5(2)^x$

GRADING NOTE y starts at 1.5 and doubles each time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q4/5

True or false?

The linear function with f(0) = 10 that decreases by 2 for each unit increase in x is $f(x) = 10(0.8)^x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q5/5

2121

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal differences give f(x) = 10 − 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 1

Write a linear function from two points or a table • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write the linear function through (0, −3) and (4, 5).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q1/5

2123

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2x − 3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 1, 2, 3 and y = 10, 7, 4.

Write the linear function.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q2/5

2124

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13 − 3x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q2/5

Which line passes through (−1, 2) and (1, 6)?

A. y = −2x + 4

B. y = 2x + 4

C. y = 2x + 2

D. y = 4x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q3/5

2125

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 4

GRADING NOTE The slope is 4 ÷ 2 = 2; 6 = 2 + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q3/5

True or false?

The line through (0, 0) and (3, 6) is y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q4/5

2126

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The slope is 2 and the y-intercept is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 1) and (2, 3) is y = 2x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q5/5

2127

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE y = 2x − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 2

Write formulas for arithmetic and geometric sequences • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A geometric sequence has a(1) = 2 and ratio 3.

Write an explicit formula.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q1/5

2129

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$2 \cdot 3^{n-1}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula for 10, 7, 4, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q2/5

2130

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

13 − 3n

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a recursive formula for 6, 3, 1.5, ...?

A. $a(n) = 6 - 3n$

B. $a(1) = 6$, $a(n) = 0.5a(n - 1)$

C. $a(1) = 6$, $a(n) = 2a(n - 1)$

D. $a(1) = 6$, $a(n) = a(n - 1) - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q3/5

2131

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a(1) = 6$, $a(n) = 0.5a(n - 1)$

GRADING NOTE Each term is half the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function on the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q4/5

2132

QUESTION ID
S56-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Equal differences between consecutive terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The geometric sequence 5, 10, 20, ... has ratio 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q5/5

2133

QUESTION ID
S56-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The ratio is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 1

Build a sinusoidal model from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

A tide has high water 10 ft and low water 2 ft, with period 12 hours and high tide at t = 0.

Write a cosine model h(t).

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q1/5

2135

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4cos(πt ÷ 6) + 6

GRADING NOTE Any equivalent form. Amplitude 4, midline 6, and 2π ÷ 12 = π ÷ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sinusoid has maximum 10 and minimum 2.

What are its amplitude and midline?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q2/5

2136

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Amplitude 4; midline y = 6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q2/5

A model y = cos(Bt) has period 12.

What is B?

A. 12

B. π ÷ 12

C. π ÷ 6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q3/5

2137

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. π ÷ 6

GRADING NOTE 2π ÷ B = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q3/5

True or false?

A sinusoidal model with maximum 8 and minimum 0 has amplitude 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q4/5

2138

QUESTION ID
S56-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE (8 − 0) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 3cos(2t) has period 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q5/5

2139

QUESTION ID
S56-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The period is 2π ÷ 2 = π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 2

Find a quadratic through three points with a matrix equation • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

ANSWER KEY Reduced copy of the question page, with its answer

$y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7).

Write the system for a, b, and c as a matrix equation $AX = B$, with X the column [a, b, c]. Give A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q1/5

2141

QUESTION ID
S56-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A has rows [0 0 1], [1 1 1], and [4 2 1]; B is the column [1, 2, 7].

GRADING NOTE Form required: A and B. Each point (x, y) gives the row $[x^2\ x\ 1]$ and the constant y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A has rows [0 0 1], [1 1 1], and [4 2 1], X is the column [a, b, c], and B is the column [1, 2, 7]. Solve AX = B to find the parabola $y = ax^2 + bx + c$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q2/5

2142

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

a = 2, b = −1, c = 1, so $y = 2x^2 - x + 1$

GRADING NOTE Form required: all three values (or the equation). c = 1, a + b = 1, and 4a + 2b = 6, so 2a = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q2/5

A parabola $y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7). It is written as $AX = B$ with X the column [a, b, c].

Which row of A comes from the point (2, 7)?

A. [4 2 1]

B. [2 4 1]

C. [4 2 7]

D. [2 2 1]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q3/5

2143

QUESTION ID

S56-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. [4 2 1]

GRADING NOTE $4a + 2b + c = 7$, so the row is [4 2 1] and the matching entry of B is 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q3/5

True or false?

Three points with different x-values that are not on one line determine exactly one parabola $y = ax^2 + bx + c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q4/5

2144

QUESTION ID
S56-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The 3-by-3 system has exactly one solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q4/5

True or false?

A parabola $y = ax^2 + bx + c$ through (0, 1), (1, 2), and (2, 7) gives a matrix equation $AX = B$ in which A has 2 rows.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q5/5

2145

QUESTION ID

S56-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each of the 3 points gives one equation, so A has 3 rows (and 3 columns).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q5/5

Observe that exponential growth eventually exceeds linear, quadratic, and polynomial growth

MAIN	Algebra I main question	**HSF-LE.A.3**
BACKWARD 1	Tell equal differences from equal factors	**HSF-LE.A.1.a**
BACKWARD 2	Compare two functions given in different forms	**8.F.A.2**
FORWARD 1	Compare the growth of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Compare logarithmic growth with other growth	**HSF-BF.B.5**

SET 57 • HSF-LE.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 100x$ and $g(x) = 2^x$.

What is the smallest whole number $x \geq 1$ for which $g(x) > f(x)$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q1/5

2148

QUESTION ID

S57-M-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $g(9) = 512 < 900 = f(9)$, and $g(10) = 1{,}024 > 1{,}000 = f(10)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q1/5

$f(x) = x^2$ and $g(x) = 1.5^x$.

Which is larger at x = 10? At x = 20? What happens for larger x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q2/5

2149

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

f at x = 10 (100 compared with about 57.7); g at x = 20 (about 3,325 compared with 400); g stays larger from then on.

GRADING NOTE Form required: all three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Plan A pays $1,000 per day. Plan B pays 1 cent on day 1, and its daily pay doubles each day.

On which day does Plan B's daily pay first exceed Plan A's?

Day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q3/5

2150

QUESTION ID
S57-M-Q3

TYPE
Short answer

CORRECT ANSWER
Day 18

GRADING NOTE Any equivalent form. Day n pays $0.01 \cdot 2^{n-1}$ dollars: day 17 pays $655.36 and day 18 pays $1,310.72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function eventually has the greatest values as x increases?

A. $y = 50x^2$

B. $y = 1.01^x$

C. $y = x^3 + 500$

D. $y = 1{,}000x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q4/5

2151

QUESTION ID

S57-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 1.01^x$

GRADING NOTE Any exponential function with base greater than 1 eventually exceeds every polynomial function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q4/5

True or false?

Because $3x^2 > 2^x$ at $x = 5$, $3x^2$ stays greater than 2^x for all $x > 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q5/5

2152

QUESTION ID

S57-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At $x = 10$, $3x^2 = 300$ and $2^x = 1{,}024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q5/5

SET 57 • HSF-LE.A.3

BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 3, 6, 12, 24.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q1/5

2154

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Exponential: y is multiplied by 2 each time.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table shows x = 0, 1, 2, 3 and y = 3, 6, 9, 12.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q2/5

2155

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Linear: y increases by 3 each time.

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function grows by equal factors over equal intervals?

A. $y = 7 + 1.1x$

B. $y = 7(1.1)^x$

C. $y = 7x$

D. $y = 1.1x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q3/5

2156

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 7(1.1)^x$

GRADING NOTE Only the exponential function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

Linear functions grow by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q4/5

2157

QUESTION ID
S57-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The difference is the slope times the interval length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q4/5

True or false?

$y = 5x^2$ grows by equal factors over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q5/5

2158

QUESTION ID
S57-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $5(2)^2 \div 5(1)^2 = 4$ but $5(3)^2 \div 5(2)^2 = 2.25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

BACKWARD 2

Compare two functions given in different forms • Grade 8 • Functions • 8.F.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Function A is y = 4x + 1. Function B has x = 0, 1, 2 and y = 3, 5, 7.

Which has the greater rate of change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q1/5

2160

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A: 4 compared with 2.

GRADING NOTE Form required: the function and both rates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q1/5

Function A is y = −x + 6. Function B is the line through (0, 2) and (2, 8).

Which has the greater y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q2/5

2161

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A: 6 compared with 2.

GRADING NOTE Form required: the function and both intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q2/5

$f(x) = 3x - 2$. Function g has $g(0) = 1$ and $g(1) = 5$, and it is linear.

Which statement is true?

A. f has the greater rate of change.

B. f has the greater initial value.

C. g has the greater rate of change and the greater initial value.

D. f and g have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q3/5

2162

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. g has the greater rate of change and the greater initial value.

GRADING NOTE g has rate 4 and initial value 1; f has rate 3 and initial value −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q3/5

True or false?

A linear function with slope 5 increases faster than one with slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q4/5

2163

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It gains 5 per unit instead of 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 2x + 10 has greater values than y = 3x for every x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q5/5

2164

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For x > 10, 3x is greater.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 1

Compare the growth of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

For which $x > 0$ is $x^3 > 100x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q1/5

2166

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x > 100

GRADING NOTE Form required: an inequality. Divide by $x^2 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which grows faster for large x, x^4 or $1{,}000x^3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q2/5

2167

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x^4: $x^4 \div (1{,}000x^3) = x \div 1{,}000$, which is greater than 1 when $x > 1{,}000$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

As $x \to \infty$, what does $f(x) = x^4 - 50x^3$ do?

A. $f(x) \to -\infty$

B. $f(x) \to -50$

C. $f(x) \to \infty$

D. $f(x) \to 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q3/5

2168

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $f(x) \to \infty$

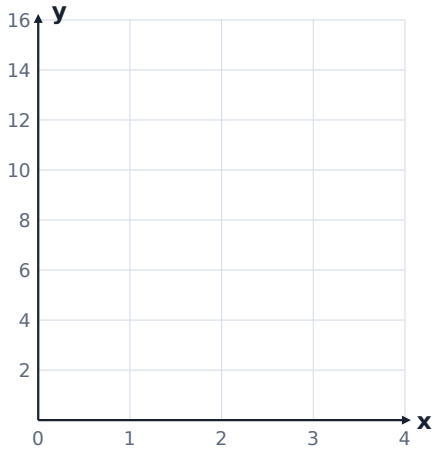
GRADING NOTE The leading term x^4 decides the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y = x^2$ and $y = 0.5x^3$ for $0 \le x \le 4$ on the same grid. Mark where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q4/5

2169

QUESTION ID

S57-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Both curves through (0, 0) and meeting again at (2, 4); $y = 0.5x^3$ is below $y = x^2$ for $0 < x < 2$ and above it for $x > 2$ (it leaves the grid near $x = 3.2$)

GRADING NOTE Any correct graph with both meeting points. $x^2 = 0.5x^3$ gives $x = 0$ or $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For every large x, $0.001x^3$ is less than $1{,}000x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q5/5

2170

QUESTION ID
S57-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE For $x > 1{,}000{,}000$, $0.001x^3$ is greater.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 2

Compare logarithmic growth with other growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Find $\log_2 1{,}024$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q1/5

2172

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $2^{10} = 1{,}024$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

By what factor must x be multiplied for log x to increase by 1? (log means base 10.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q2/5

2173

QUESTION ID
S57-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE Any equivalent form. log(10x) = 1 + log x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q2/5

Which is greatest when x = 1,000,000? (log means base 10; ln means base e.)

A. $\sqrt{x}$

B. $\log_2 x$

C. ln x

D. log x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q3/5

2174

QUESTION ID
S57-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. $\sqrt{x}$

GRADING NOTE $\sqrt{x} = 1{,}000$; $\log x = 6$; $\ln x \approx 13.8$; $\log_2 x \approx 19.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

Eventually, log x grows more slowly than any linear function with positive slope. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q4/5

2175

QUESTION ID
S57-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE log x is the inverse of the fast-growing 10^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

$\log_2 x = 100$ when $x = 200$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q5/5

2176

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = 2^{100}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q5/5

Interpret the parameters of a linear or exponential function in context

MAIN	Algebra I main question	**HSF-LE.B.5**
BACKWARD 1	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
BACKWARD 2	Interpret the rate and initial value of a linear function	**8.F.B.4**
FORWARD 1	Interpret the parameters of a sinusoidal model	**HSF-TF.B.5**
FORWARD 2	Interpret the mean and standard deviation of a normal model — n…	**HSS-ID.A.4**

SET 58 • HSF-LE.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$C(m) = 0.35m + 40$ is the cost in dollars of renting a car for one day and driving m miles.

What do 0.35 and 40 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q1/5

2179

QUESTION ID
S58-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.35: the cost per mile ($0.35); 40: the daily fee ($40) charged even for 0 miles.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q1/5

$P(t) = 1{,}200(1.04)^t$ is a town's population t years after 2020.

What do 1,200 and 1.04 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q2/5

2180

QUESTION ID
S58-M-Q2

TYPE
Short answer

CORRECT ANSWER
1,200: the population in 2020; 1.04: the population grows 4% each year.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q2/5

$A(t) = 80(0.5)^{t/6}$ is the milligrams of a medicine left in the body after t hours.

What do 80, 0.5, and 6 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q3/5

2181

QUESTION ID

S58-M-Q3

TYPE

Short answer

CORRECT ANSWER

80 mg is the starting dose; half of the medicine remains after every 6 hours (a half-life of 6 hours).

GRADING NOTE Form required: all three meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

V(t) = 18,000 − 1,500t is a machine's value in dollars after t years.

What does −1,500 mean?

A. The machine loses $1,500 of value each year.

B. The machine loses 1,500% of its value each year.

C. The machine was bought for $1,500.

D. The machine is worth $18,000 after 1,500 years.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q4/5

2182

QUESTION ID

S58-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The machine loses $1,500 of value each year.

GRADING NOTE The slope is the change in value per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q4/5

True or false?

In $y = 250(0.92)^t$, the value decreases by 92% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S58-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q5/5

2183

QUESTION ID

S58-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 92% remains each year, so it decreases by 8%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S58-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 1

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges 75 + 60h dollars for h hours of work.

What does 75 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q1/5

2185

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A $75 fee charged for every visit, before any hours of work.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In $3{,}000(1.05)^t$, what does the factor 1.05 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q2/5

2186

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The amount grows 5% for each unit of t.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q2/5

A box with a square base of side s and height h has surface area $2s^2 + 4sh$.

What does 4sh represent?

A. The volume

B. The area of the four side faces

C. The perimeter of the base

D. The area of the top and bottom

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q3/5

2187

QUESTION ID

S58-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The area of the four side faces

GRADING NOTE Each side face is s by h, and there are four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

In 500 − 20w, the term −20w shows a decrease of 20 for each unit of w.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q4/5

2188

QUESTION ID

S58-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The coefficient −20 is the change per unit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In 1.5x + 7, the coefficient of x is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q5/5

2189

QUESTION ID
S58-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The coefficient of x is 1.5; 7 is the constant term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 2

Interpret the rate and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

Cost y = 12x + 30 for x hours of bike rental.

What is the initial value, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q1/5

2191

GRADING NOTE Form required: the value and its meaning.

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30: a $30 fee before any hours are counted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q1/5

The water in a tank is $y = 500 - 25t$ gallons after t minutes.

What does -25 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q2/5

2192

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The tank loses 25 gallons each minute.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q2/5

A plant's height is a linear function of time. It is 8 cm on day 0 and 20 cm on day 4.

What is the rate of change?

A. 5 cm per day

B. 20 cm per day

C. 8 cm per day

D. 3 cm per day

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q3/5

2193

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 cm per day

GRADING NOTE (20 − 8) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q3/5

True or false?

In y = 40x + 100, 100 is the value of y when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q4/5

2194

QUESTION ID

S58-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In d = 65t, 65 is the starting distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q5/5

2195

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 65 is the rate (distance per unit of time); the starting distance is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q5/5

2195

SET 58 • HSF-LE.B.5

FORWARD 1

Interpret the parameters of a sinusoidal model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What do 15 and 10 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q1/5

2197

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15 °C is the average (midline) temperature; the temperature rises and falls 10 °C above and below it.

GRADING NOTE Form required: both meanings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What is the period of T, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q2/5

2198

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

24 hours: the temperature pattern repeats every day.

GRADING NOTE Form required: the period and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q2/5

$h(t) = 1.5\cos(\pi t \div 6) + 3$ is the depth of water in meters.

What is the greatest depth?

A. 4.5 m

B. 3 m

C. 6 m

D. 1.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q3/5

2199

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.5 m

GRADING NOTE 3 + 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In y = A sin(Bx) + D, D gives the midline.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q4/5

2200

QUESTION ID

S58-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is centered on y = D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q4/5

True or false?

The amplitude of $y = 1.5\cos(\pi t \div 6) + 3$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q5/5

2201

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The amplitude is 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

FORWARD 2

Interpret the mean and standard deviation of a normal model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Interpreting the parameters of a function in context leads to interpreting the parameters of a normal model.

ANSWER KEY Reduced copy of the question page, with its answer

Battery life is normal with mean 400 hours and standard deviation 20 hours.

What does the 20 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q1/5 • NEAREST RELATED

2203

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

It measures spread: battery lives typically differ from the mean by roughly 20 hours (about 68% are within 20 hours of 400).

GRADING NOTE Any equivalent description of the standard deviation as a typical (root-mean-square) distance from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q1/5 • NEAREST RELATED

2203

ANSWER KEY Reduced copy of the question page, with its answer

Battery life is normal with mean 400 hours and standard deviation 20 hours.

About what percent of batteries last between 360 and 440 hours?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q2/5 • NEAREST RELATED

2204

QUESTION ID
S58-F2-Q2

TYPE
Short answer

CORRECT ANSWER
About 95%

GRADING NOTE Any equivalent form. That is within 2 standard deviations of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

Which change makes a normal curve wider?

A. Increasing the standard deviation

B. Increasing the mean

C. Decreasing the standard deviation

D. Decreasing the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q3/5 • NEAREST RELATED

2205

QUESTION ID
S58-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. Increasing the standard deviation

GRADING NOTE The standard deviation measures spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q3/5 • NEAREST RELATED

2205

True or false?

Changing the mean of a normal model shifts the curve left or right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q4/5 • NEAREST RELATED

2206

QUESTION ID

S58-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The mean is its center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a normal model with mean 50 and standard deviation 5, the value 60 is 10 standard deviations above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q5/5 • NEAREST RELATED

2207

QUESTION ID

S58-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(60 - 50) \div 5 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 59 • HSS-ID.A.1 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Represent data with dot plots, histograms, and box plots

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display numerical data in plots	**6.SP.B.4**
BACKWARD 2	Compute measures of center and variation	**6.SP.B.5.c**
FORWARD 1	Decide whether a normal model fits a distribution	**HSS-ID.A.4**
FORWARD 2	Decide whether results are consistent with a model by simulation…	**HSS-IC.A.2**

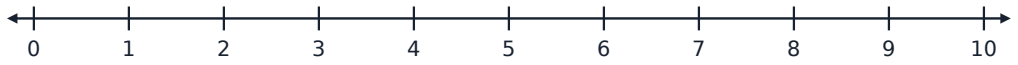
SET 59 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a dot plot of the numbers of books read by 10 students:

2, 3, 3, 4, 4, 4, 5, 5, 6, 9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q1/5

2210

QUESTION ID
S59-M-Q1

TYPE
Drawing / plotting

CORRECT ANSWER
One dot above 2, two above 3, three above 4, two above 5, one above 6, and one above 9

GRADING NOTE Any correct dot plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the five-number summary of 2, 3, 3, 4, 4, 4, 5, 5, 6, 9. Use the median of each half (lower five and upper five values) as Q1 and Q3.

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q2/5

2211

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

Minimum 2, Q1 3, median 4, Q3 5, maximum 9

GRADING NOTE Form required: all five values. The median is the mean of the 5th and 6th values; Q1 and Q3 are the medians of the lower and upper five values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q2/5

For the data 2, 3, 3, 4, 4, 4, 5, 5, 6, 9 (Q1 and Q3 are the medians of the lower five and upper five values), use the 1.5 × IQR rule to decide whether 9 is an outlier.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q3/5

2212

QUESTION ID

S59-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. IQR = 5 − 3 = 2, and Q3 + 1.5(2) = 8. Since 9 > 8, it is an outlier.

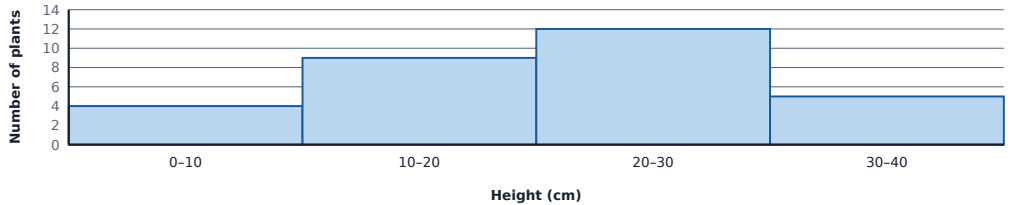
GRADING NOTE Form required: the answer with the fence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q3/5

The histogram shows the heights of 30 plants. Each bar includes its lower edge but not its upper edge.

How many plants are at least 20 cm tall?



A. 21 **B.** 17 **C.** 5 **D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q4/5

2213

QUESTION ID

S59-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17

GRADING NOTE The bars for 20–30 and 30–40 have 12 + 5 = 17 plants.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q4/5

True or false?

A box plot shows the mean of a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q5/5

2214

QUESTION ID
S59-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A box plot shows the five-number summary; the mean is not shown.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 1

Display numerical data in plots • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Which display shows every data value, a dot plot or a box plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q1/5

2216

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A dot plot

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Make a dot plot of 1, 2, 2, 3, 3, 3, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q2/5

2217

GRADING NOTE Any correct dot plot.

QUESTION ID

S59-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

One dot above 1, two above 2, three above 3, and one above 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which display groups data values into intervals?

A. Box plot

B. Dot plot

C. Histogram

D. Ordered list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q3/5

2218

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Histogram

GRADING NOTE A histogram counts the values in each interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q3/5

True or false?

A box plot is made from the five-number summary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q4/5

2219

QUESTION ID

S59-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Minimum, Q1, median, Q3, and maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A histogram shows each individual data value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q5/5

2220

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows only how many values fall in each interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 2

Compute measures of center and variation • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the median of 3, 8, 5, 10, 4.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q1/5

2222

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. In order: 3, 4, 5, 8, 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 3, 8, 5, 10, 4.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q2/5

2223

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. 30 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Q1 = 12 and Q3 = 20.

What is the interquartile range?

A. 32

B. 8

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q3/5

2224

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

GRADING NOTE 20 − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The range of 2, 7, 9, 15 is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q4/5

2225

QUESTION ID

S59-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 15 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q4/5

True or false?

The median of 1, 2, 3, 100 is 26.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q5/5

2226

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median is (2 + 3) ÷ 2 = 2.5; 26.5 is the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 1

Decide whether a normal model fits a distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A data set is strongly skewed to the right.

Is a normal model appropriate? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q1/5

2228

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No. A normal distribution is symmetric, and this one is not.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Scores are normal with mean 75 and standard deviation 8.

What score is 1 standard deviation above the mean?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q2/5

2229

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

83

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Values are normal with mean 20 and standard deviation 4.

About what percent are between 12 and 28?

A. 99.7%

B. 50%

C. 68%

D. 95%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q3/5

2230

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 95%

GRADING NOTE 12 and 28 are 2 standard deviations from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q3/5

True or false?

In a normal distribution, the mean and the median are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q4/5

2231

QUESTION ID

S59-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The distribution is symmetric about its mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

About 50% of a normal distribution lies within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q5/5

2232

QUESTION ID

S59-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE About 68% does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 2

Decide whether results are consistent with a model by simulation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Simulation results are summarized with the data displays of the MAIN set and then compared with a model.

ANSWER KEY Reduced copy of the question page, with its answer

In 100 simulated sets of 10 fair coin flips, 8 or more heads happened in 5 sets.

Estimate the probability of 8 or more heads in 10 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q1/5 • NEAREST RELATED

2234

QUESTION ID
S59-F2-Q1

TYPE
Short answer

CORRECT ANSWER
0.05

GRADING NOTE Any equivalent form, such as 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q1/5 • NEAREST RELATED

A spinner model says red comes up 25% of the time. In 200 spins, red came up 52 times.

What proportion is that? Is it consistent with the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q2/5 • NEAREST RELATED

2235

QUESTION ID
S59-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.26; yes, it is close to 0.25.

GRADING NOTE Form required: the proportion and a judgment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A model says a coin lands heads 50% of the time.

Which result most strongly questions the model?

A. 52 heads in 100 flips

B. 6 heads in 10 flips

C. 95 heads in 100 flips

D. 48 heads in 100 flips

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q3/5 • NEAREST RELATED

2236

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 95 heads in 100 flips

GRADING NOTE 95 of 100 is very far from 50 and almost never happens with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A result that rarely happens in simulations of a model is evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q4/5 • NEAREST RELATED

2237

QUESTION ID
S59-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It suggests the model may be wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If a coin lands heads 6 times in 10 flips, the coin must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q5/5 • NEAREST RELATED

2238

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6 of 10 happens often with a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 59 • FORWARD 2 • Q5/5 • NEAREST RELATED

Make histograms from data

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display data in a histogram	**6.SP.B.4**
BACKWARD 2	Report the number of observations	**6.SP.B.5.a**
FORWARD 1	Use a histogram's shape to decide whether a normal model fits	**HSS-ID.A.4**
FORWARD 2	Use a histogram of simulated results to judge a model	**HSS-IC.A.2**

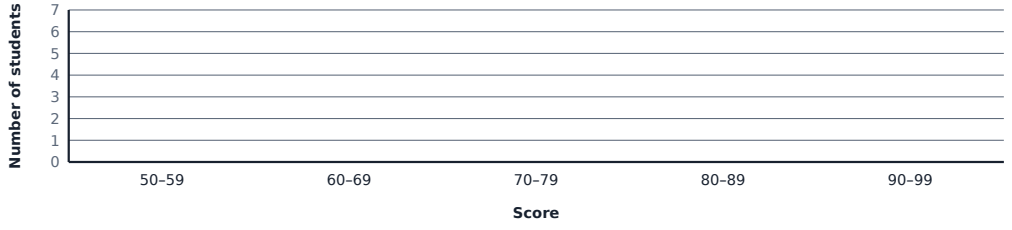
SET 76 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a histogram of these 20 test scores. Use the intervals 50–59, 60–69, 70–79, 80–89, and 90–99.

52, 58, 61, 64, 67, 69, 70, 72, 74, 75, 77, 78, 81, 83, 85, 86, 88, 91, 94, 97



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q1/5

2241

QUESTION ID
S76-M-Q1

TYPE
Drawing / plotting

CORRECT ANSWER
Bars of heights 2, 4, 6, 5, and 3

GRADING NOTE Any correct histogram; the bar heights must add to 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using the intervals 50–59, 60–69, 70–79, 80–89, and 90–99 for the scores 52, 58, 61, 64, 67, 69, 70, 72, 74, 75, 77, 78, 81, 83, 85, 86, 88, 91, 94, 97:

how many scores are in the 70–79 interval, and which interval has the tallest bar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q2/5

2242

QUESTION ID
S76-M-Q2

TYPE
Short answer

CORRECT ANSWER
6 scores; the 70–79 interval is tallest.

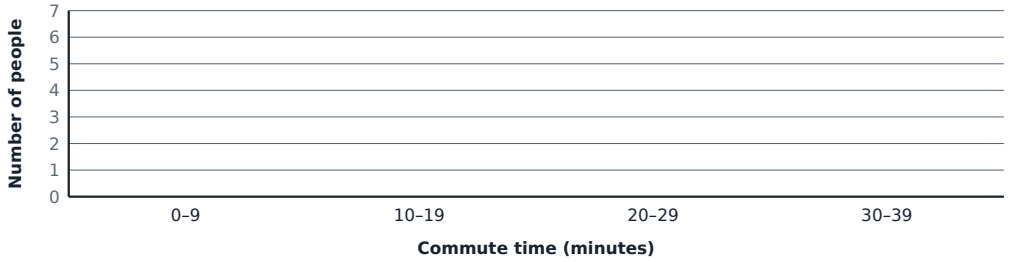
GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q2/5

Make a histogram of these 16 commute times, in minutes. Use the intervals 0–9, 10–19, 20–29, and 30–39.

4, 8, 12, 14, 15, 18, 19, 21, 22, 22, 25, 28, 29, 33, 36, 38



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q3/5

2243

QUESTION ID
S76-M-Q3

TYPE
Drawing / plotting

CORRECT ANSWER
Bars of heights 2, 5, 6, and 3

GRADING NOTE Any correct histogram; the bar heights must add to 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram uses the intervals 10–19 and 20–29 for whole-number data.

In which bar is the value 20 counted?

A. 20–29

B. 10–19

C. Both

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q4/5

2244

QUESTION ID

S76-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 20–29

GRADING NOTE Each value belongs to exactly one interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q4/5

True or false?

Using wider intervals for the same data gives a histogram with more bars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q5/5

2245

QUESTION ID

S76-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each wider interval covers more values, so fewer bars are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q5/5

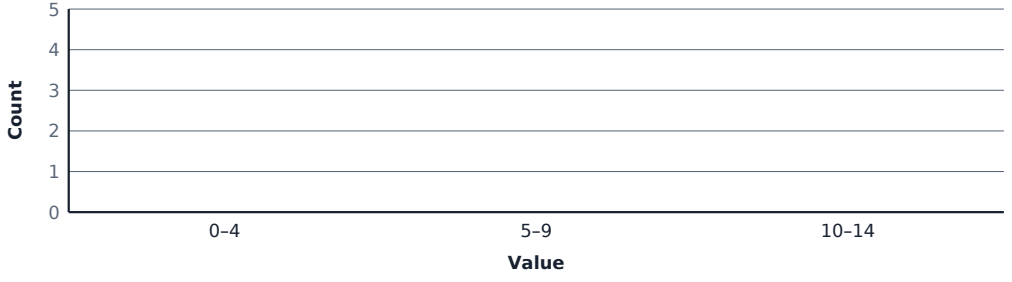
SET 76 • HSS-ID.A.1

BACKWARD 1

Display data in a histogram • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Make a histogram of 1, 3, 4, 6, 7, 8, 9, 12 with the intervals 0–4, 5–9, and 10–14.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q1/5

2247

QUESTION ID

S76-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Bars of heights 3, 4, and 1

GRADING NOTE Any correct histogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram has bars of heights 3, 5, and 2. How many data values does it show?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q2/5

2248

QUESTION ID

S76-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q2/5

2248

Which data are best shown in a histogram?

A. The heights of 200 students

B. Each student's favorite color

C. One test score

D. The names of 5 cities

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q3/5

2249

QUESTION ID

S76-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The heights of 200 students

GRADING NOTE A histogram shows numerical data grouped into intervals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a histogram, the height of a bar shows how many values are in that interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q4/5

2250

QUESTION ID

S76-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Bar heights are counts (frequencies).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A histogram is the best display for categories such as favorite colors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q5/5

2251

QUESTION ID

S76-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Categories are shown with a bar graph, which has gaps between bars.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSS-ID.A.1

BACKWARD 2

Report the number of observations • Grade 6 • Statistics & Probability • 6.SP.B.5.a

ANSWER KEY Reduced copy of the question page, with its answer

How many observations are in the data set 4, 7, 7, 9, 10, 12?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q1/5

2253

QUESTION ID

S76-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram has bars of heights 2, 4, 6, 5, and 3. How many observations does it show?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q2/5

2254

QUESTION ID

S76-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A survey recorded the number of pets owned by each of 25 students. How many observations are there?

A. 1

B. 25

C. 100

D. The total number of pets

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q3/5

2255

QUESTION ID

S76-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25

GRADING NOTE One observation per student.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q3/5

True or false?

The number of observations is the number of data values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q4/5

2256

QUESTION ID

S76-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each value is one observation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q4/5

True or false?

If 8 students each report 2 siblings, there are 16 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q5/5

2257

QUESTION ID

S76-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are 8 observations (each is the value 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSS-ID.A.1

FORWARD 1

Use a histogram's shape to decide whether a normal model fits • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A histogram of 1,000 adult heights is single-peaked and symmetric, with mean 170 cm and standard deviation 8 cm. About how many heights are between 162 cm and 178 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q1/5

2259

QUESTION ID
S76-F1-Q1

TYPE
Short answer

CORRECT ANSWER
About 680

GRADING NOTE Any equivalent form. That interval is within 1 standard deviation of the mean: about 68% of 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram of household incomes is strongly skewed right. Would a normal model give good estimates of percents? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q2/5

2260

QUESTION ID
S76-F1-Q2

TYPE
Short answer

CORRECT ANSWER
No. A normal model is symmetric, and the data are strongly skewed.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q2/5

Which histogram shape suggests that a normal model may fit?

A. Two separate peaks

B. Flat (uniform)

C. Single-peaked and symmetric

D. Strongly skewed to the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q3/5

2261

QUESTION ID

S76-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Single-peaked and symmetric

GRADING NOTE A normal curve is one symmetric bell.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q3/5

True or false?

In a normal model, about 95% of values are within 2 standard deviations of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q4/5

2262

QUESTION ID

S76-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is part of the 68-95-99.7 rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Any histogram with one peak is normal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q5/5

2263

QUESTION ID
S76-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A single peak can still be skewed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSS-ID.A.1

FORWARD 2

Use a histogram of simulated results to judge a model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

100 simulated trials of 20 fair coin flips give a histogram of the number of heads centered near 10. What does the histogram show?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q1/5

2265

QUESTION ID
S76-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The distribution of the number of heads when the coin is fair: which counts are typical and which are rare.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q1/5

In that simulation, 2 of the 100 trials had 15 or more heads. Estimate the probability of 15 or more heads in 20 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q2/5

2266

QUESTION ID
S76-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.02

GRADING NOTE Any equivalent form, such as 2%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner is claimed to land on blue 50% of the time. In 50 spins it landed on blue 12 times. In 200 simulated sets of 50 spins with a 50% chance, 12 or fewer blue never happened.

What do you conclude?

A. Nothing can be concluded.

B. The result supports the 50% claim.

C. The spinner lands on blue exactly 24% of the time.

D. The result is not consistent with the 50% claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q3/5

2267

QUESTION ID
S76-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. The result is not consistent with the 50% claim.

GRADING NOTE The observed result is far outside what the model usually produces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q3/5

True or false?

A simulation repeats a random process many times to show which results are typical under a model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q4/5

2268

QUESTION ID
S76-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is the purpose of the simulation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q4/5

True or false?

If a result never happens in 200 simulated trials, it is impossible under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q5/5

2269

QUESTION ID
S76-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is rare, not impossible; more trials might produce it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.2 • Set 76 • FORWARD 2 • Q5/5

Make box plots from data

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Draw and read box plots	**6.SP.B.4**
BACKWARD 2	Find medians, quartiles, and ranges	**6.SP.B.5.c**
FORWARD 1	Estimate the quartiles of a normal distribution	**HSS-ID.A.4**
FORWARD 2	Compare two treatments with box plots and a randomization test	**HSS-IC.B.5**

SET 77 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a box plot of 3, 5, 6, 7, 8, 9, 10, 12, 15. (Use the median of each half, not counting the middle value, as Q1 and Q3.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q1/5

2272

QUESTION ID
S77-M-Q1

TYPE
Drawing / plotting

CORRECT ANSWER
Whiskers from 3 to 5.5 and from 11 to 15; a box from Q1 = 5.5 to Q3 = 11 with the median line at 8

GRADING NOTE Any correct box plot. Lower half 3, 5, 6, 7 gives Q1 = 5.5; upper half 9, 10, 12, 15 gives Q3 = 11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the five-number summary of 12, 15, 15, 18, 20, 22, 25, 30. (Q1 and Q3 are the medians of the lower four and upper four values.)

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q2/5

2273

QUESTION ID

S77-M-Q2

TYPE

Short answer

CORRECT ANSWER

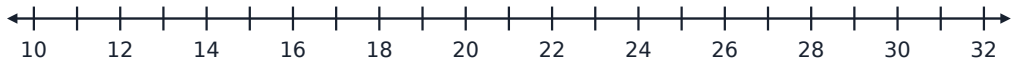
Minimum 12, Q1 15, median 19, Q3 23.5, maximum 30

GRADING NOTE Form required: all five values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q2/5

Make a box plot of 12, 15, 15, 18, 20, 22, 25, 30. (Q1 and Q3 are the medians of the lower four and upper four values.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q3/5

2274

QUESTION ID
S77-M-Q3

TYPE
Drawing / plotting

CORRECT ANSWER

Whiskers from 12 to 15 and from 23.5 to 30; a box from 15 to 23.5 with the median line at 19

GRADING NOTE Any correct box plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q3/5

In a box plot, what does the length of the box show?

A. The interquartile range

B. The mean

C. The range

D. The median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q4/5

2275

QUESTION ID

S77-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The interquartile range

GRADING NOTE The box runs from Q1 to Q3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q4/5

True or false?

In a box plot whose whiskers reach the minimum and maximum, each whisker covers about a quarter of the data values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q5/5

2276

QUESTION ID

S77-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The quartiles split the ordered data into four parts of about equal size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSS-ID.A.1

BACKWARD 1

Draw and read box plots • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Draw a box plot with minimum 2, Q1 4, median 5, Q3 8, and maximum 10.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q1/5

2278

QUESTION ID

S77-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Whiskers from 2 to 4 and from 8 to 10; a box from 4 to 8 with a line at 5

GRADING NOTE Any correct box plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A box plot has Q1 = 20 and Q3 = 35. What is the interquartile range?

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q2/5

2279

QUESTION ID

S77-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which part of a box plot marks the median?

A. The line inside the box

B. The left end of the box

C. The right whisker

D. The left whisker

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q3/5

2280

QUESTION ID

S77-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The line inside the box

GRADING NOTE The median divides the box.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q3/5

2280

True or false?

A box plot shows the minimum and the maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q4/5

2281

QUESTION ID

S77-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are the ends of the whiskers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A box plot shows how many data values there are.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q5/5

2282

QUESTION ID

S77-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows the five-number summary, not the count.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSS-ID.A.1

BACKWARD 2

Find medians, quartiles, and ranges • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the median of 4, 9, 2, 7, 5, 8.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q1/5

2284

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. In order: 2, 4, 5, 7, 8, 9; (5 + 7) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The ordered data are 2, 4, 5, 7, 8, 9. Find Q1, the median of the lower half 2, 4, 5.

Q1 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q2/5

2285

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of 14, 3, 9, 21?

A. 21

B. 18

C. 7

D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q3/5

2286

QUESTION ID

S77-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18

GRADING NOTE 21 − 3 = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The interquartile range is Q3 − Q1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q4/5

2287

QUESTION ID

S77-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The median of 1, 3, 100 is 34.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q5/5

2288

QUESTION ID

S77-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median is the middle value, 3; 34 is the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSS-ID.A.1

FORWARD 1

Estimate the quartiles of a normal distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

For a normal distribution, the quartiles are about $\mu - 0.67\sigma$ and $\mu + 0.67\sigma$. Estimate Q1 and Q3 for mean 100 and standard deviation 15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q1/5

2290

QUESTION ID

S77-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Q1 ≈ 90 and Q3 ≈ 110

GRADING NOTE Form required: both values (about 89.95 and 110.05 are also accepted). 0.67 × 15 ≈ 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using those quartiles, estimate the interquartile range of that normal distribution.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q2/5

2291

QUESTION ID

S77-F1-Q2

TYPE

Short answer

CORRECT ANSWER

About 20

GRADING NOTE Any equivalent form. 110 − 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a normal distribution, about what percent of values are between Q1 and Q3?

A. 25%

B. 95%

C. 50%

D. 68%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q3/5

2292

QUESTION ID
S77-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. 50%

GRADING NOTE Q1 to Q3 always holds the middle half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q3/5

2292

True or false?

A box plot of data from a normal distribution is roughly symmetric about the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q4/5

2293

QUESTION ID
S77-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A normal distribution is symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For normal data, the median is far from the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q5/5

2294

QUESTION ID

S77-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For a normal distribution they are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q5**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSS-ID.A.1

FORWARD 2

Compare two treatments with box plots and a randomization test • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Students were randomly assigned to a new study method (group A) or the usual method (group B). Box plots show a median of 78 for A and 70 for B, with similar spreads. In 1,000 random regroupings, a difference in means at least as large as observed happened 30 times.

Using a 5% threshold, is the difference significant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q1/5

2296

QUESTION ID
S77-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Yes.
30 ÷ 1,000 = 3%, which is below 5%.

GRADING NOTE Form required: the decision with the proportion compared with 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q1/5

Why is it useful to draw side-by-side box plots of the two groups before running the test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q2/5

2297

QUESTION ID
S77-F2-Q2

TYPE
Short answer

CORRECT ANSWER
They show the centers, spreads, overlap, and any outliers of the two groups at a glance.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q2/5

In another randomized comparison, random regroupings matched or beat the observed difference 22% of the time.

Using a 5% threshold, what is the conclusion?

A. The difference is significant.

B. The methods are proved to have the same effect.

C. The new method is proved to work.

D. The difference is not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q3/5

2298

QUESTION ID

S77-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The difference is not significant.

GRADING NOTE 22% is above 5%, so chance could easily explain the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q3/5

True or false?

In a randomized comparison, a difference that rarely appears in random regroupings is evidence of a treatment effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q4/5

2299

QUESTION ID
S77-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Chance alone rarely produces it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q4/5

True or false?

Overlapping box plots prove that two treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q5/5

2300

QUESTION ID
S77-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Overlap shows variation; it does not prove the effects are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q5/5

2300

SET 60 • HSS-ID.A.2 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Compare the center and spread of two or more data sets

MAIN	Algebra I main question	**HSS-ID.A.2**
BACKWARD 1	Tell measures of center from measures of variation	**6.SP.A.3**
BACKWARD 2	Compare two distributions using center and variability	**7.SP.B.3**
FORWARD 1	Use simulation to decide whether a difference between treatment…	**HSS-IC.B.5**
FORWARD 2	Evaluate reports based on data	**HSS-IC.B.6**

SET 60 • HSS-ID.A.2

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Class A scores: 70, 75, 80, 85, 90. Class B scores: 60, 70, 80, 90, 100.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q1/5

2303

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

Both have mean and median 80. Class B is more spread out: its range is 40 compared with 20 (and its standard deviation is larger).

GRADING NOTE Form required: a comparison of center and of spread, with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the population standard deviation (divide by n) of 2, 4, 4, 4, 5, 5, 7, 9.

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q2/5

2304

QUESTION ID

S60-M-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. The mean is 5; the squared deviations add to 32; 32 ÷ 8 = 4, and √4 = 2. (Dividing by n − 1 gives about 2.14.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-M-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q2/5

Store X's daily sales have median 50 and IQR 10. Store Y's have median 45 and IQR 25.

Which store typically sells more? Which is more consistent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q3/5

2305

QUESTION ID

S60-M-Q3

TYPE

Short answer

CORRECT ANSWER

Store X on both: its median is higher, and its IQR is smaller.

GRADING NOTE Form required: both answers with reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q3/5

Which measure of spread is best for skewed data with outliers?

A. Range

B. Standard deviation

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q4/5

2306

QUESTION ID

S60-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Interquartile range

GRADING NOTE The IQR uses the middle half of the data, so outliers barely affect it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q5/5

2307

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Counterexample: 70, 75, 80, 85, 90 and 60, 70, 80, 90, 100 both have mean 80, but the second set is more spread out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 1

Tell measures of center from measures of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

Which describes a typical value of a data set, the median or the IQR?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q1/5

2309

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The median

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of the data set 4, 4, 4, 4?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q2/5

2310

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form. There is no variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q2/5

Which is a measure of variation?

A. Median

B. Mode

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q3/5

2311

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Interquartile range

GRADING NOTE The median, the mean, and the mode are measures of center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q3/5

True or false?

A measure of center summarizes all of the values with a single number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q4/5

2312

QUESTION ID

S60-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, the mean or the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The range is a measure of center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q5/5

2313

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The range measures variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 2

Compare two distributions using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Two classes have mean scores 78 and 84, and each has MAD 3.

How many MADs apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q1/5

2315

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (84 − 78) ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Team A's mean height is 170 cm and Team B's is 180 cm; each has MAD 5 cm.

Express the difference of the means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q2/5

2316

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 MADs

GRADING NOTE Any equivalent form. 10 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q2/5

Two means differ by 12, and each data set has MAD 4.

How many MADs is the difference?

A. 48

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q3/5

2317

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE 12 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

When two distributions have little variability, a small difference in their centers is easier to notice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q4/5

2318

QUESTION ID

S60-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The data overlap less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If two data sets overlap, their means must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q5/5

2319

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Overlapping sets can have different means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 1

Use simulation to decide whether a difference between treatments is significant • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Plants were randomly assigned to a fertilizer group or a control group with no fertilizer. The fertilizer group averaged 24 cm and the control group 20 cm. To test chance alone, the same heights were randomly regrouped 1,000 times, and a difference of 4 cm or more (first group minus second) happened 12 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q1/5

2321

QUESTION ID
S60-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. Only about 1.2% of the regroupings gave a difference this large, which is below 5%, so chance alone is an unlikely explanation.

GRADING NOTE Form required: the decision with the simulated proportion (12 ÷ 1,000 = 1.2%) compared with 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Why are subjects randomly assigned to treatments in an experiment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q2/5

2322

QUESTION ID

S60-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Random assignment tends to balance other variables between the groups, so a large difference can be credited to the treatment rather than to how the groups were formed.

GRADING NOTE Any equivalent explanation. Random assignment balances other variables on average; it does not make the groups identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In a regrouping simulation, a difference as large as the observed one happened 400 times out of 1,000.

What does that suggest?

A. The treatment definitely works.

B. The difference could easily be due to chance.

C. The treatment definitely does not work.

D. The experiment had no control group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q3/5

2323

QUESTION ID
S60-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. The difference could easily be due to chance.

GRADING NOTE 40% is not rare.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q3/5

True or false?

In a randomized experiment, a small simulated probability for the observed difference is evidence that the treatment had an effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q4/5

2324

QUESTION ID
S60-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Random assignment makes chance the main other explanation, and chance alone rarely produces the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q4/5

True or false?

In an experiment, the control group receives the new treatment being tested.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q5/5

2325

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The control group is the comparison group: it gets no treatment, a placebo, or a standard treatment, not the new one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 2

Evaluate reports based on data • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

An ad says "9 out of 10 dentists recommend Brand X," based on 10 dentists chosen by Brand X.

What is the flaw?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q1/5

2327

QUESTION ID
S60-F2-Q1

TYPE
Short answer

CORRECT ANSWER
The sample is tiny and was chosen by the company, so it is likely biased.

GRADING NOTE Any equivalent explanation that names size or bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An online poll of a website's visitors finds 80% support a policy.

Can the result be applied to all adults? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q2/5

2328

QUESTION ID
S60-F2-Q2

TYPE
Short answer

CORRECT ANSWER

No. The respondents chose themselves and are only that site's visitors, so they do not represent all adults.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sample is most likely to represent a school's students?

A. The first 50 students to arrive

B. The 50 students in the chess club

C. A random sample of 50 students from the school roster

D. 50 volunteers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q3/5

2329

QUESTION ID
S60-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. A random sample of 50 students from the school roster

GRADING NOTE Only random selection gives every student the same chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q3/5

True or false?

A larger random sample generally gives a smaller margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q4/5

2330

QUESTION ID
S60-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Estimates from larger random samples vary less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A survey with many volunteers is free of bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q5/5

2331

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers may differ from the population, whatever their number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.3 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Interpret differences in shape, center, and spread, accounting for outliers

MAIN	Algebra I main question	**HSS-ID.A.3**
BACKWARD 1	Describe a distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Choose measures of center and spread that fit the data	**6.SP.B.5.d**
FORWARD 1	Use z-scores to judge how unusual a value is	**HSS-ID.A.4**
FORWARD 2	Distinguish a population parameter from a sample statistic — nea…	**HSS-IC.A.1**

SET 61 • HSS-ID.A.3

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

Find the mean and the median. Which better describes a typical salary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q1/5

2334

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

Mean ≈ 74.29 thousand; median 45 thousand. The median is better: the one salary of 250 pulls the mean up.

GRADING NOTE Form required: both values (the mean to the nearest hundredth or as 520/7) and the choice with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

If the salary of 250 is removed, what happens to the mean and to the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q2/5

2335

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

The mean drops from about 74.29 to 45; the median stays 45.

GRADING NOTE Form required: both changes. 270 ÷ 6 = 45, and the middle values are 45 and 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q2/5

A histogram of wait times has a long tail on the right.

Describe its shape and the usual relationship between its mean and median.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q3/5

2336

QUESTION ID

S61-M-Q3

TYPE

Short answer

CORRECT ANSWER

Skewed right; the mean is usually greater than the median, because the long right tail pulls the mean up.

GRADING NOTE Form required: the shape and the usual relationship. This is a typical tendency, not a rule that holds for every data set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which data set has the largest standard deviation?

A. 28, 29, 30, 31, 32

B. 25, 30, 30, 30, 35

C. 10, 20, 30, 40, 50

D. 30, 30, 30, 30, 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q4/5

2337

QUESTION ID

S61-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10, 20, 30, 40, 50

GRADING NOTE All four have mean 30; the values 10, 20, 30, 40, 50 are spread farthest from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q4/5

True or false?

An outlier affects the median more than the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q5/5

2338

QUESTION ID
S61-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The mean uses every value; the median depends only on the middle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 1

Describe a distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Data: 1, 2, 2, 3, 3, 3, 4, 4, 5.

Describe the shape of the distribution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q1/5

2340

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric, centered at 3

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value in 2, 3, 3, 3, 4, 15 is unusual?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q2/5

2341

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A dot plot has most values on the left and a few values far to the right.

What is its shape?

A. Skewed right

B. Symmetric

C. Uniform

D. Skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q3/5

2342

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Skewed right

GRADING NOTE The tail points to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A distribution can be described by its center, its spread, and its shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q4/5

2343

QUESTION ID

S61-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE These are the three main features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A symmetric distribution must have an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q5/5

2344

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Many symmetric distributions have no outliers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 2

Choose measures of center and spread that fit the data • Grade 6 • Statistics & Probability • 6.SP.B.5.d

ANSWER KEY Reduced copy of the question page, with its answer

For data with an extreme outlier, should you report the mean or the median as the center? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q1/5

2346

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The median, because one extreme value barely changes it.

GRADING NOTE Form required: the choice and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For symmetric data with no outliers, which measures of center and spread are usually reported?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q2/5

2347

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The mean and the mean absolute deviation

GRADING NOTE Any equivalent form. (The mean and the standard deviation are also accepted.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Home prices in a town include one mansion.

Which measure best describes a typical price?

A. The mean

B. The median

C. The maximum

D. The range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q3/5

2348

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The median

GRADING NOTE The mansion pulls the mean up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

The median is not changed much by a single extreme value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q4/5

2349

QUESTION ID

S61-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It depends on the middle of the ordered data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q4/5

True or false?

The IQR changes greatly when the largest value is made even larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q5/5

2350

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The IQR depends on Q1 and Q3, which do not change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.3

FORWARD 1

Use z-scores to judge how unusual a value is • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A distribution has mean 50 and standard deviation 4.

What is the z-score of 58?

z = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

2352

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (58 − 50) ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A score of 92 on a test with mean 80 and standard deviation 6, or a score of 85 on a test with mean 70 and standard deviation 10: which is better relative to its test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

2353

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

92: its z-score is 2, compared with 1.5.

GRADING NOTE Form required: the score and both z-scores.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

In a normal distribution, which z-score is the most unusual?

A. −3.1

B. −0.5

C. 1.2

D. 2.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

2354

QUESTION ID
S61-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. −3.1

GRADING NOTE It is the farthest from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A z-score of 0 means the value equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

2355

QUESTION ID
S61-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE It is 0 standard deviations from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A z-score of −2 is above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

2356

QUESTION ID

S61-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Negative z-scores are below the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

FORWARD 2

Distinguish a population parameter from a sample statistic • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Centers and spreads computed from data are sample statistics that estimate population parameters.

ANSWER KEY Reduced copy of the question page, with its answer

What is the difference between a population and a sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q1/5 • NEAREST RELATED

2358

QUESTION ID
S61-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A population is the whole group of interest; a sample is part of it that is actually studied.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 200 students has a mean of 7.1 hours of sleep.

What does 7.1 estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q2/5 • NEAREST RELATED

2359

QUESTION ID
S61-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The mean hours of sleep of all the students in the population.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

A parameter describes

A. a graph

B. one individual

C. a population

D. a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q3/5 • NEAREST RELATED

2360

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. a population

GRADING NOTE A statistic describes a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Different random samples from the same population can give different sample means.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q4/5 • NEAREST RELATED

2361

QUESTION ID

S61-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A sample statistic always equals the population parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q5/5 • NEAREST RELATED

2362

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is only an estimate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 61 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 62 • HSS-ID.B.5 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Summarize categorical data in two-way frequency tables and interpret relative frequencies

MAIN	Algebra I main question	**HSS-ID.B.5**
BACKWARD 1	Read two-way tables for association	**8.SP.A.4**
BACKWARD 2	Write a part of a whole as a percent	**7.RP.A.3**
FORWARD 1	Use a two-way table to find conditional probabilities and test inde…	**HSS-CP.A.4**
FORWARD 2	Explain conditional probability and independence in everyday lang..	**HSS-CP.A.5**

SET 62 • HSS-ID.B.5

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a survey of 200 students.

What percent of all the students play a sport?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q1/5

2365

QUESTION ID

S62-M-Q1

TYPE

Short answer

CORRECT ANSWER

52.5%

GRADING NOTE Any equivalent form. 105 ÷ 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q1/5

What percent of Grade 10 students play a sport? What percent of Grade 9 students do?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q2/5

2366

QUESTION ID

S62-M-Q2

TYPE

Short answer

CORRECT ANSWER

60% of Grade 10; 45% of Grade 9

GRADING NOTE Form required: both conditional relative frequencies. 60 ÷ 100 and 45 ÷ 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q2/5

Is there an association between grade and playing a sport? Explain.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q3/5

2367

QUESTION ID

S62-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. 60% of Grade 10 students play a sport compared with 45% of Grade 9 students.

GRADING NOTE Form required: the answer and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the joint relative frequency of "Grade 9 and does not play a sport"?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. 50% **B.** 57.9% **C.** 55% **D.** 27.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q4/5

2368

QUESTION ID

S62-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 27.5%

GRADING NOTE 55 out of all 200 students. (55% is the share of Grade 9 students; about 57.9% is the share of non-players.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q4/5

True or false?

Of the students who play a sport, more than half are in Grade 10.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q5/5

2369

QUESTION ID

S62-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 60 ÷ 105 ≈ 57%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 1

Read two-way tables for association • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

12 of 20 boys and 9 of 15 girls in a class like math.

How many students in all like math?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q1/5

2371

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

12 of 20 boys and 9 of 15 girls in a class like math.

Is there an association between gender and liking math? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q2/5

2372

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. 12 ÷ 20 = 60% and 9 ÷ 15 = 60%: the same share.

GRADING NOTE Form required: the answer with both relative frequencies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the row and column totals of a two-way table called?

A. Joint frequencies

B. Conditional frequencies

C. Marginal frequencies

D. Outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q3/5

2373

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Marginal frequencies

GRADING NOTE They are written in the margins of the table.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q3/5

True or false?

A two-way table shows data on two categorical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q4/5

2374

QUESTION ID

S62-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One variable for the rows and one for the columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a two-way table, every row must have the same total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q5/5

2375

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Row totals can be different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 2

Write a part of a whole as a percent • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write 45 out of 100 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q1/5

2377

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 60 out of 105 as a percent, rounded to the nearest tenth.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q2/5

2378

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

57.1%

GRADING NOTE Form required: rounded to the nearest tenth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

27 is what percent of 60?

A. 45%

B. 60%

C. 33%

D. 27%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q3/5

2379

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 45%

GRADING NOTE 27 ÷ 60 = 0.45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

55 out of 200 is 27.5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q4/5

2380

QUESTION ID

S62-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 55 ÷ 200 = 0.275.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

40 out of 95 is 40%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q5/5

2381

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 40 ÷ 95 ≈ 42.1%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 1

Use a two-way table to find conditional probabilities and test independence • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A student from the table is chosen at random.

Find P(plays a sport | Grade 10).

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q1/5

2383

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE Any equivalent form. 60 ÷ 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are "Grade 10" and "plays a sport" independent? Explain with probabilities.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q2/5

2384

QUESTION ID
S62-F1-Q2

TYPE
Short answer

CORRECT ANSWER
No. P(plays a sport) = 0.525, but P(plays a sport | Grade 10) = 0.6.

GRADING NOTE Form required: the answer with both probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is P(Grade 9 | plays a sport)?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. About 0.43

B. 0.225

C. 0.525

D. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q3/5

2385

QUESTION ID
S62-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. About 0.43

GRADING NOTE 45 ÷ 105 ≈ 0.43.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q3/5

True or false?

If P(B) > 0, events A and B are independent exactly when P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q4/5

2386

QUESTION ID

S62-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Knowing B does not change the probability of A. P(B) > 0 is needed for P(A | B) to be defined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(A | B) always equals P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q5/5

2387

QUESTION ID

S62-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE In the table, P(sport | Grade 10) = 0.6 but P(Grade 10 | sport) ≈ 0.57.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 2

Explain conditional probability and independence in everyday language • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Explain in everyday language what P(late | rain) > P(late) means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q1/5

2389

QUESTION ID
S62-F2-Q1

TYPE
Short answer

CORRECT ANSWER
A person is more likely to be late on a rainy day than on a typical day.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A and B are independent, P(A) = 0.3, and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q2/5

2390

QUESTION ID
S62-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.15

GRADING NOTE Any equivalent form. 0.3 × 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of events is most likely independent?

A. Height and shoe size

B. Studying and passing a test

C. Flipping heads and rolling a 6

D. Rain and carrying an umbrella

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q3/5

2391

QUESTION ID
S62-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Flipping heads and rolling a 6

GRADING NOTE A coin and a number cube do not affect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q3/5

True or false?

If A and B are independent, then P(A and B) = P(A) · P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q4/5

2392

QUESTION ID
S62-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the multiplication rule for independent events.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If P(A | B) = 0.8 and P(A) = 0.4, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q5/5

2393

QUESTION ID
S62-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Knowing B doubles the probability of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear, quadratic, or exponential function to data and use it to solve problems

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Describe patterns in scatter plots	**8.SP.A.1**
BACKWARD 2	Write linear and exponential functions through given points	**HSF-LE.A.2**
FORWARD 1	Use logarithms with an exponential model fitted to data	**HSF-LE.A.4**
FORWARD 2	Fit a sinusoidal model to periodic data	**HSF-TF.B.5**

SET 63 • HSS-ID.B.6.a

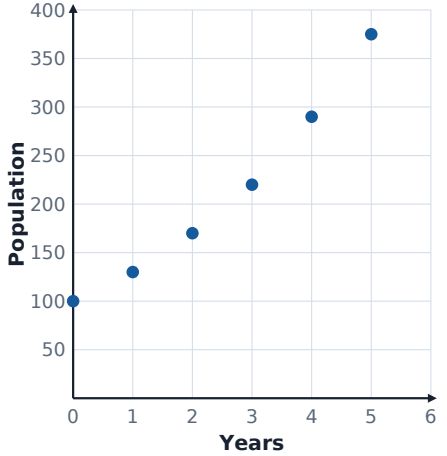
MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

ANSWER KEY Reduced copy of the question page, with its answer

The scatter plot shows a population over 5 years.

Which type of function fits best? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q1/5

2396

QUESTION ID

S63-M-Q1

TYPE

Short answer

CORRECT ANSWER

Exponential: the population is multiplied by about 1.3 each year (130 ÷ 100, 170 ÷ 130, ...).

GRADING NOTE Form required: the type and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

An exponential model for the data is $P(t) = 100(1.3)^t$.

Use it to predict the population at t = 8, to the nearest whole number.

P(8) ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q2/5

2397

QUESTION ID

S63-M-Q2

TYPE

Short answer

CORRECT ANSWER

816

GRADING NOTE Form required: rounded to the nearest whole number. $100(1.3)^8 \approx 815.7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Data: x = 0, 1, 2, 3, 4 and y = 2, 5, 10, 17, 26.

Which quadratic function fits these data exactly? Write it.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q3/5

2398

QUESTION ID

S63-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x + 2$

GRADING NOTE Any equivalent form, such as $(x + 1)^2 + 1$. The second differences are all 2, so the x^2 coefficient is 1; y(0) = 2 gives the constant 2, and y(1) = 5 gives the x coefficient 2. Check: y(4) = 16 + 8 + 2 = 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q3/5

A scatter plot rises and then falls in a symmetric curve.

Which model fits best?

A. Quadratic

B. Exponential growth

C. Exponential decay

D. Linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q4/5

2399

QUESTION ID

S63-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Quadratic

GRADING NOTE A parabola rises and then falls symmetrically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q4/5

True or false?

Using a model to predict far outside the range of the data is always reliable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q5/5

2400

QUESTION ID

S63-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Extrapolation assumes the pattern continues, which it may not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q5/5